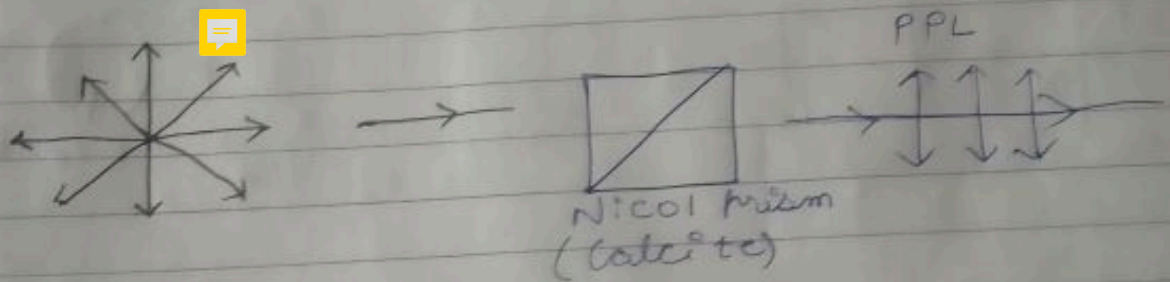
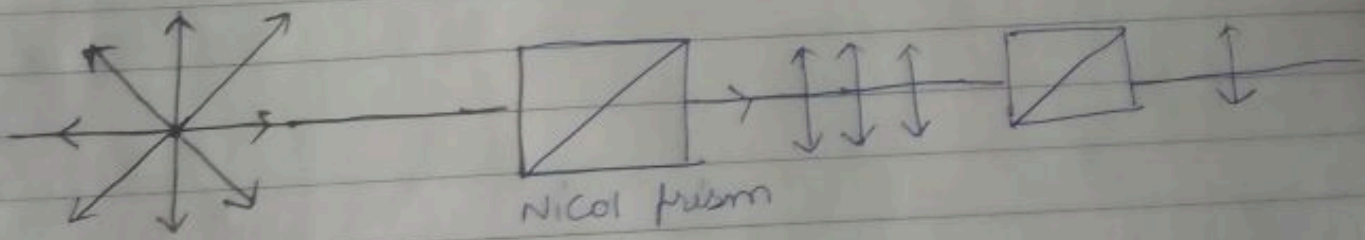
OPTICAL ISOMERISM

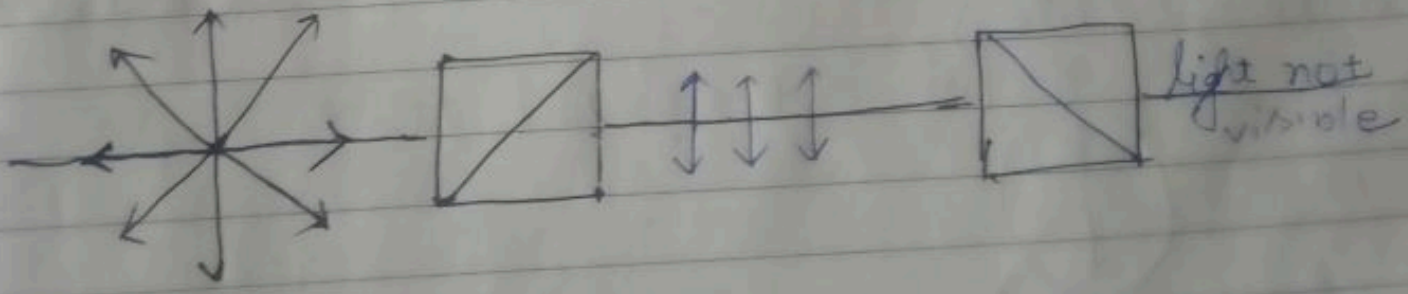
1 Optical activity :- Compounds which can rotate vibrational plane of plane polarised light (PPL) are called optical active compounds and this phenomenon is called Optical activity

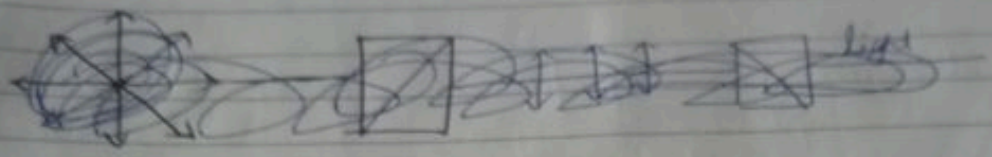
Stereoisomers which have different optical activity are called optical isomers.

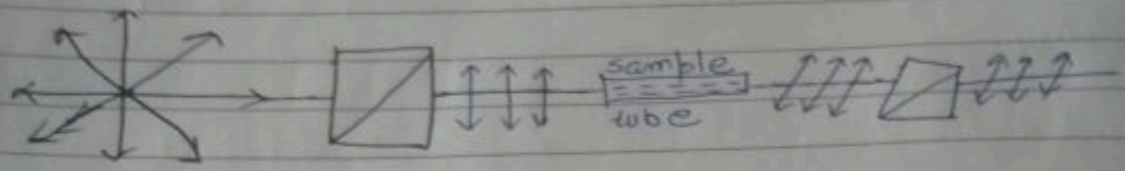
(II) Plane polarised light :- Light wave whose vibration occurs only in single plane to perpendicular w.r.t propagation light.





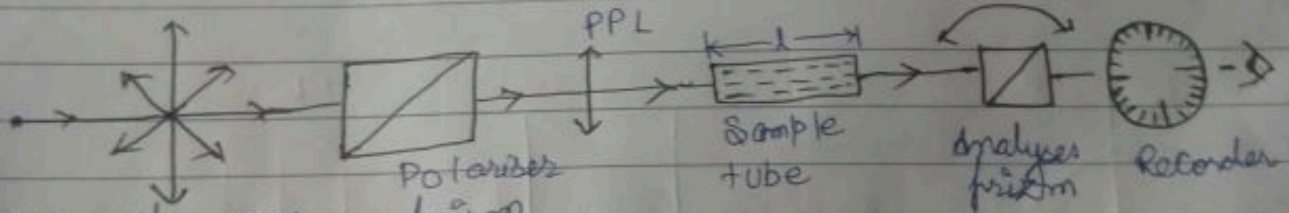






II) Polarimeter :-

By using this instrument we can measure the value of observed rotation of vibrational plane of PPL in presence of optical active compound.

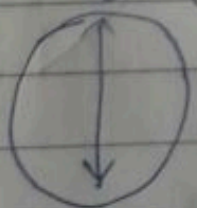


Source of mono-chromatic light

↳ all rays are of same wavelength

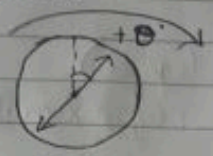
(IV) Observations on polarimeter :- for optical inactive compounds (OIC)

$\theta = 0^\circ$ θ = observed rotation



For optical active compound (OAC)

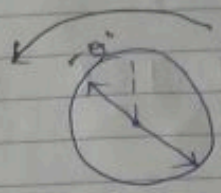
for clockwise rotation



Dextrorotatory Comp.

(d) or (+)

for anticlockwise rotation



Laevorotatory comp.

l or (-)

Dextrorotatory Comp. - Compounds which can rotate vibrational plane of ppl in clockwise direc^{n}.

Laevorotatory Comp. :- Compounds which can rotate vibrational plane of ppl in anticlockwise direc^{n}.

⇒ The magnitude of optical rotation depends upon following factors :-

1) Nature of substance
2) length of sample tube
3) Concen. of solution
4) wavelength of light (usually Na-D lamp - 589 nm)
5) Solvent
6) Temp

v) Specific rotation :- $[\alpha]$

$\theta \propto l$

$\theta \propto c$

$\theta \propto cl$

$$[\alpha]_D^{t=25^\circ C} = \frac{\theta}{cl}$$

$\theta = [\alpha] cl$

θ = obs. rotation (degree)

c = Concentration of sample (in g/ml)

l = length of sample tube (in dm)

1 dm = 10 cm

θ = observed rotation (degree)

c = Concentration of sample (in g/ml)

l = length of sample tube (in dm)

1 dm = 10 cm

Q.1 For a solution of an optical active compound the value of optical rotation is 13.4° when conc. of solution is 2g in 10ml and length of sample tube is 25 cm. find specific rotation.

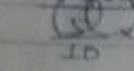
$$\alpha = \frac{13.4^\circ}{\frac{0.2 \times 25}{10}} = \frac{13.4}{\frac{50}{100}} = \frac{134 \times 100}{50}$$

$$= 268$$

Q. Identify the value of specific rotation of a solution of 1M solution of 2-butanol

length of sample tube = 5 cm

obs. rotation = +7.4°

⇒ 

$C_4H_{10}O$

$= 48 + 10 + 16$

$= 74$

$C = 1M = 1 mol/lit$

$= 74g / 1000 ml$

$l = \frac{5}{10} dm$

$\alpha = \frac{\theta}{C \cdot l} = \frac{+7.4}{\frac{7.4}{1000} \times \frac{5}{10}} = \frac{1000}{5} = 200°C$

VI) Chiral Carbon = Asymmetric Carbon

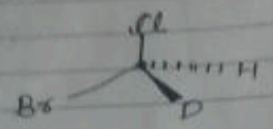
Four different group containing tetrahedral Carbon is called chiral Carbon.

Note 1. only one chiral atom containing compounds are always optically active.

Note 2. When more than one chiral atoms are present :

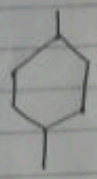
- with dissimilar ends → always optically active
- with similar ends → may be optical active / may not be active (consider 3d structure)

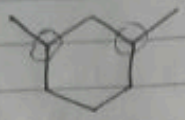
8. Identify no. of chiral carbons.

eg 1. 

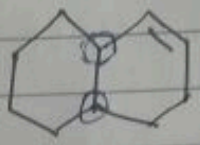
(one) [Optically Active Compound (OAC)]

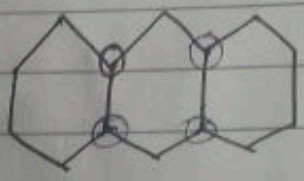
eg 2.  (Two) (OAC)

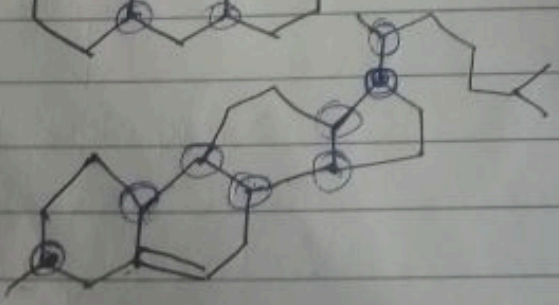
eg 3.  (Zero)

eg 4)  (Two)

eg 5)  (Zero)

eg 6)  (One) (Two) [OAC]

eg 7)  (Four)

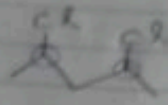


(One) [OAC]

eg 8) (Six) (eight)

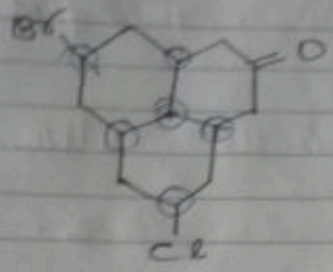
(Seven)

eg 9)



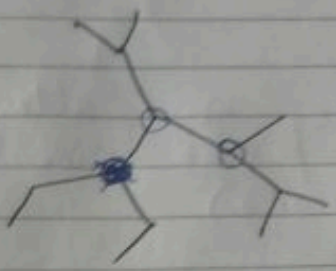
(Trans)

eg 10)



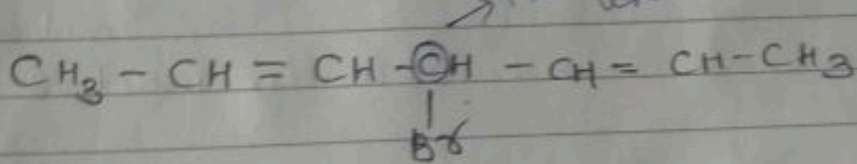
(~~Trans~~)
(Mix) [DAC]

eg 11



(~~Trans~~) [~~DAC~~]
(Trans)

eg 12)



✗ cis
✗ trans
✓ cis

cis [Not chiral]
trans
trans [chiral]

VII) Condition for optical activity :-

Optical active compounds are not superimposable on its mirror image structure. This property is called chirality

-> All chiral molecules are optically active.

~~-> All chiral molecules are optical active.~~

-> Chirality is necessary and sufficient condition for optical activity.

i) centre of Symm., plane of Sym. and alternating axis of symmetry are not present in optical active compounds (chiral molecule)

out of these 3 elements of Symm. if any one element of Symm. will present in molecule then compound will optically inactive.

Elements of Symmetry

A) axis of Symmetry (C)
B) plane of Symmetry (σ)
C) centre of Symmetry (i)
d) Alternating axis of Symmetry (S)

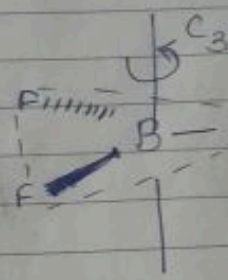
} useful for identification of optical active comp.

A) Axis of Symmetry (C_n)

It is an axis of Symmetry
It is an imaginary axis which passed through a molecule when we rotate this molecule around the axis by 360° then we obtain identical arrangement (more than one) to ~~to~~ original one

as i)

BF_3



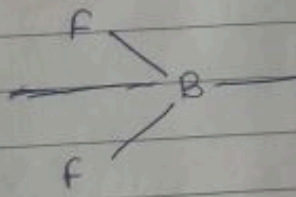
=> one C_3 axis

board ⊥ plane ⊥ ~~to~~

=> three C_2 axis

=> ∞ C_1 axis

Board ⊥ plane पर

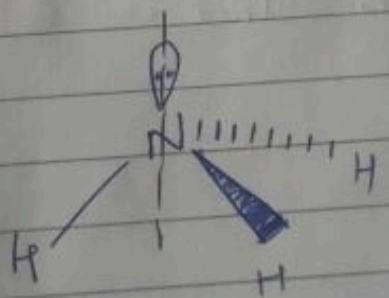


$= C_{\frac{360}{180}}$

$n = \frac{360}{180}$

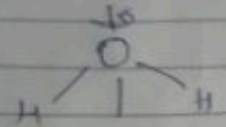
ये molecule में C_1 axis होता है।

eg 3)



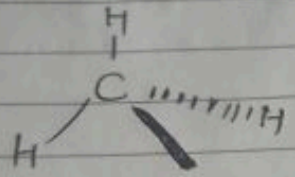
one C_3 axis

eg 3>



One C_2 axis

eg 4> CH_4



Bond angle = 109.5°
Bond angles = 6
C_3 axis = 4
C_2 axis = 3

Find C_2 axis

A B C D E F G H I J K L

$\checkmark C_2$ = A B C D E H I K M N O S T U V W

$\times C_2$ = F G J L P Q R

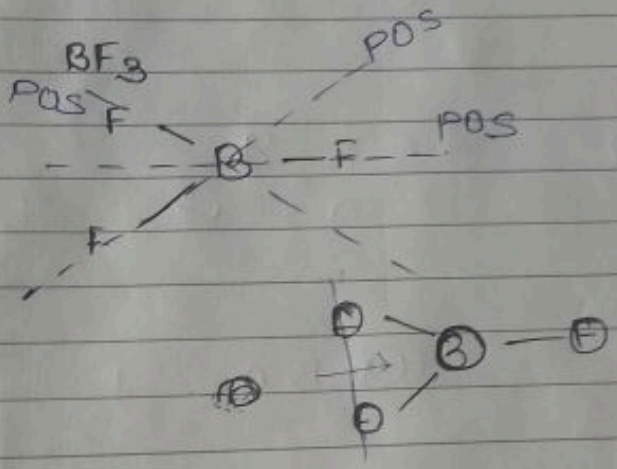
B> Plane of symmetry :

It is an imaginary plane which bisects the molecule / object in two equal half parts and these both half parts should be mirror image of each other.

c) Centre of symmetry = Point of symmetry = Inversion centre :-

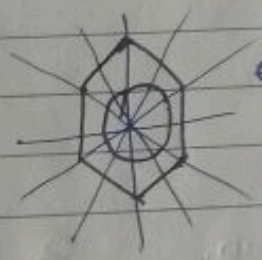
It is an imaginary point in the centre of molecule from which if line is drawn to the direction of a particular group and it is extended in opposite direction, it meets two similar image groups, on equal distance from centre of molecule.

eg 1)



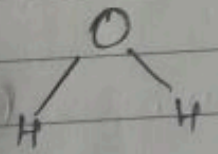
P.
P·O·S = 3 + 1 = 4
C·O·S = ✗

eg 2)



POS = 6 + 1 = 7
COS = ✓

eg 3)



POS = 2

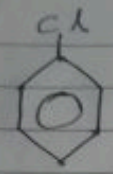
eg 4)

O = C = O

POS = ∞
COS = ✓

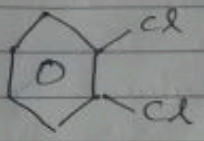
C·O·S हमेशा एक molecule में एक ही होता है

eg 5)



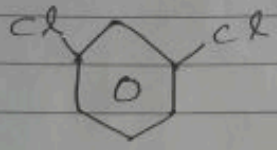
P·O·S = 2

eg 6)



P·O·S = 2

eg 7)



P·O·S = 2

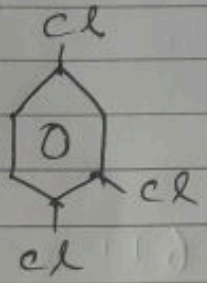
eg 8)



P·O·S = 3

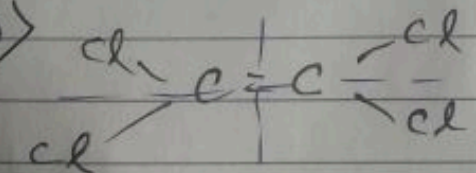
C·O·S = ✓

eg 9)



P·OS = 1

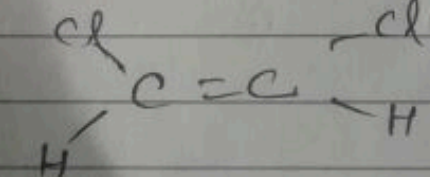
eg 10)



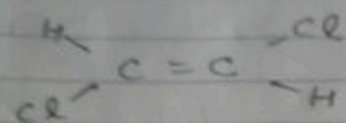
P·O·S = 3

C·O·S = ✓

eg 11)

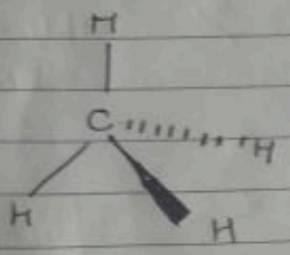


P·O·S = 2

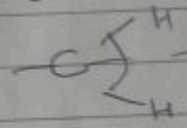
eg 12) 

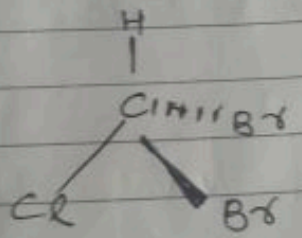
C.O.S = ✓
P.O.S = ✓

Note: All planer molecules have minimum one ~~symmetry~~ plane of symmetry (always) so all planer molecules are optically inactive (achiral).

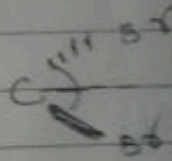
eg 13) 

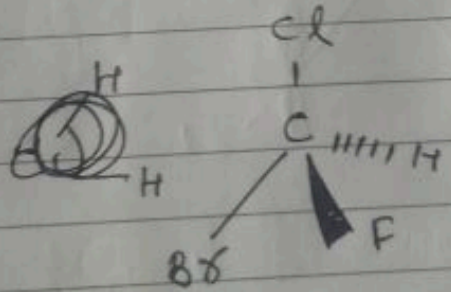
P.O.S = 6
C.O.S = X



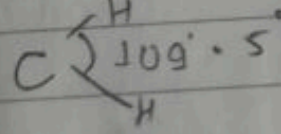
eg 14) 

P.O.S = 1
C.O.S = X



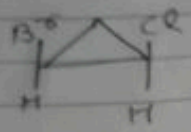
eg 15) 

P.O.S = X
C.O.S = X



6 Bond angle in CH_4

eg 16)

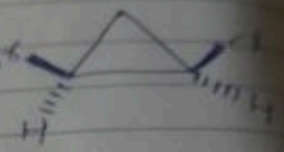


ring is in 1st of page

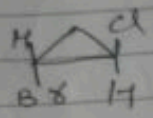
P.O.S = ✗

C.O.S = ✗

Or



eg 17)



P.O.S = ✗

C.O.S = ✗

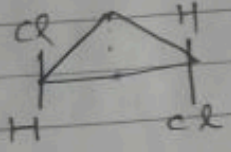
eg 18)



P.O.S = ✓ (1)

C.O.S = ✗ optically achisal molecule inactive

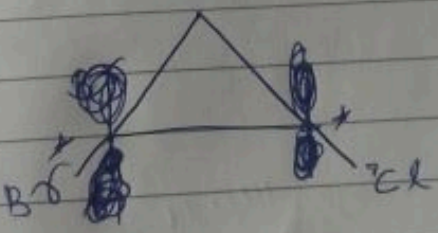
eg 19)



P.O.S = ✗

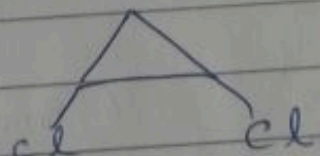
C.O.S = ✗ optically active chisal molecule

eg 16 & eg 17 are cis and trans of



dissimilar ends so optically active

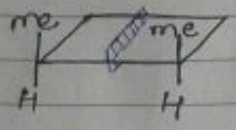
eg 18 & eg 19 are cis and trans of



Similar ends so no Comment without ed structure

VIII) elements

eg 20)

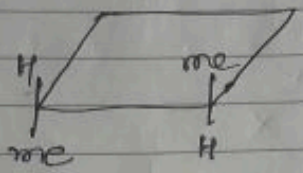


cis

C.O.S = ✗
P.O.S = ⊥ } cis

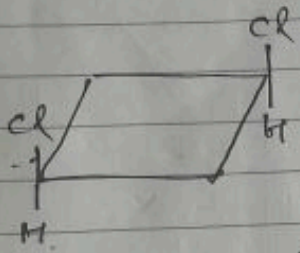


eg 21



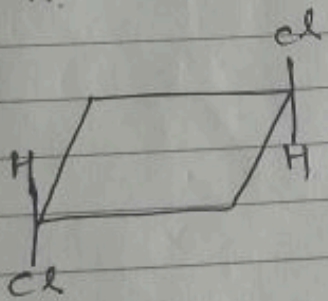
P.O.S = ✗
C.O.S = ✗ } (O.A.C) or chiral comp

eg 22)

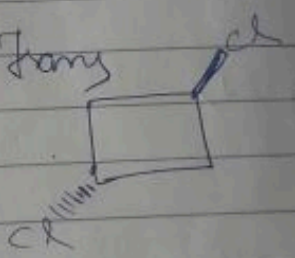


P.O.S = 2
C.O.S = ✗ } cis

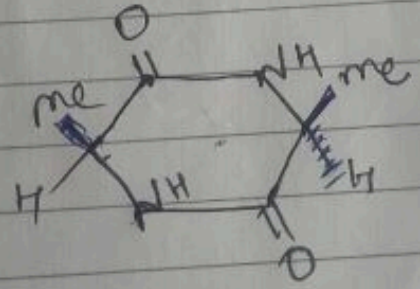
eg 23)



C.O.S = ✓
P.O.S = ⊥ } trans

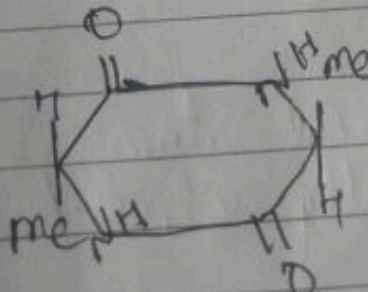


eg 24)



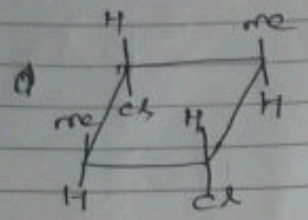
P.O.S = ✗
C.O.S = ✗

eg 25)



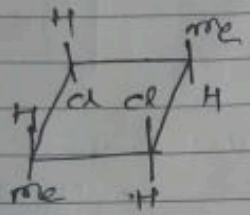
P.O.S = ✗
C.O.S = ✓

eg 26)



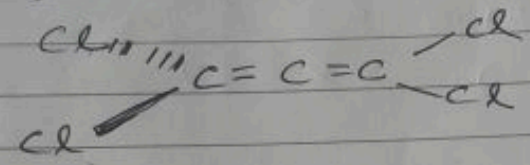
C.O.S = X
P.O.S = 2

eg 27)



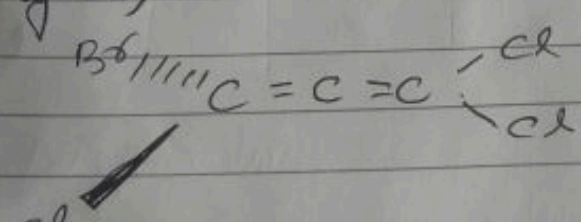
C.O.S = ✓
P.O.S = X

eg 28)



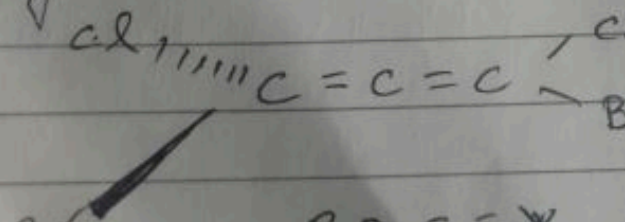
P.O.S = 2
C.O.S = X

eg 29)



P.O.S = 1
C.O.S = X

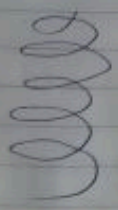
eg 30)



P.O.S = X
C.O.S = X

⇒ No chiral atom
⇒ optically active (chiral molecule)

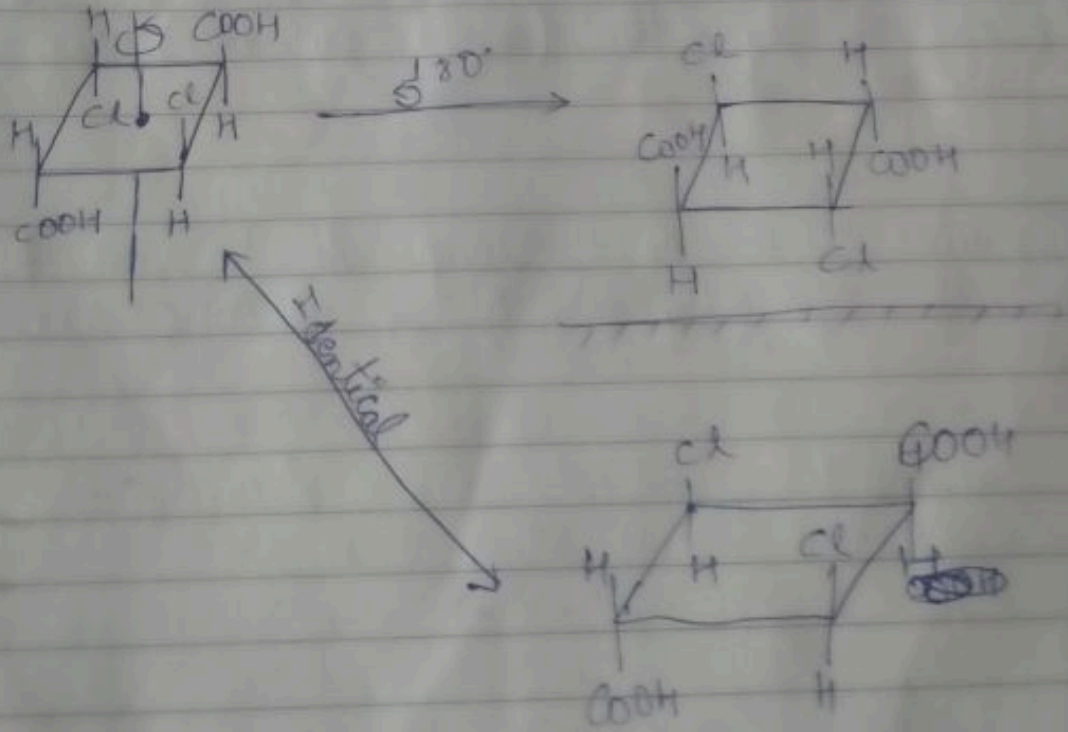
Eg 3.)



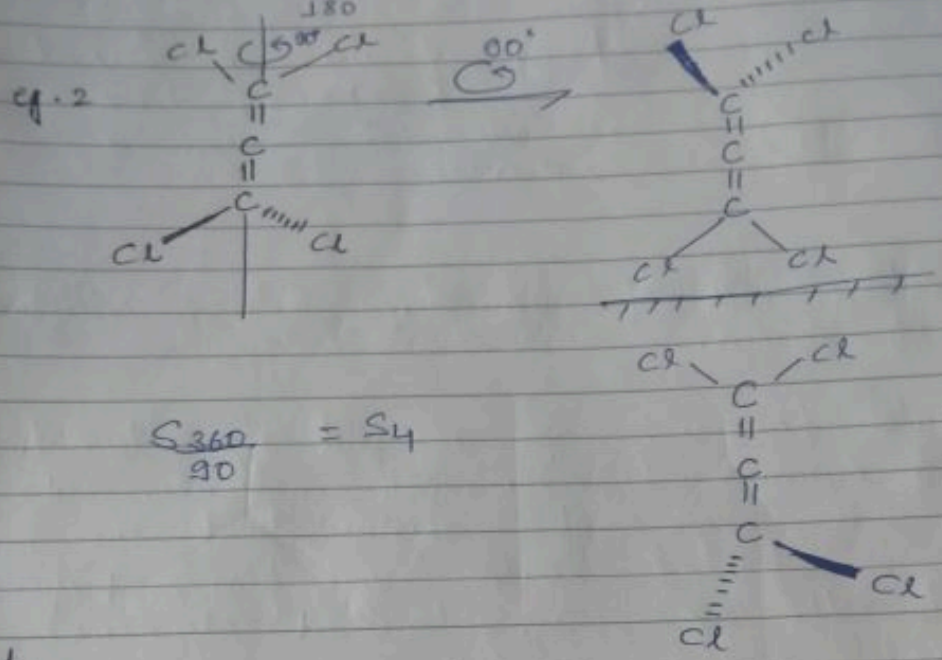
Spring is also a chiral compound bcz it has no C.O.S & P.O.S

D) Alternating axis of symmetry = Rotation reflection axis of symm.

A molecule possesses an n-fold of alternating axis of symmetry (S_n). when molecule rotated through an angle of $\frac{360°}{n}$ about the axis and then reflected across a plane perpendicular to the axis, an identical structure is obtained.



$S_n = S_{\frac{360}{180}} = S_2$



Eg. 2

$S_{\frac{360}{90}} = S_4$

Note:

⇒ Asymmetric compounds :- Compounds in which all elements of symmetries (except C_1 axis) are absent.

Dissymmetric compounds :- Compounds in which plane of symmetry, C.O.S, A.A.OS are absent.

⇒ All asymmetric molecules are dissymmetric but its reverse is not always true.

VVS

VIII Some important definitions related to optical isomerism :-

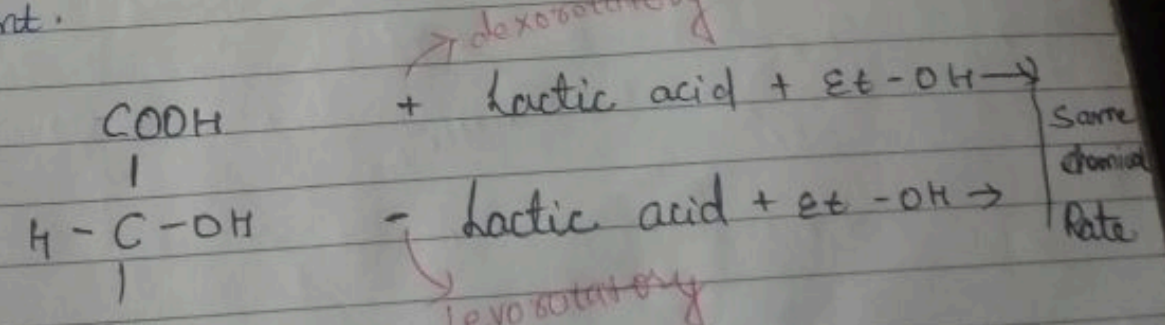
[A] Enantiomers :- Stereoisomers which have mirror image of each other but they are not superimpose on each other.

→ Enantiomers are always optically active

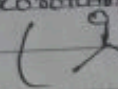
→ Enantiomers have identical physical properties like melting point, Boiling point, solubility, density etc. except the direction of rotation of light (PPL).

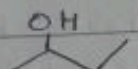
→ Both enantiomers can rotate the vibrational plane of PPL with same magnitude in opposite direction (↺ or ↻)

Enantiomers have identical chemical properties with optical inactive reagent but they have different chemical property with optical active reagent.



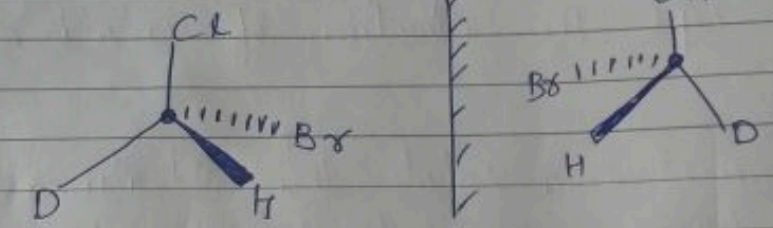
+ Lactic acid + dextrorotatory alcohol → diff



- Lactic acid + d  → diff

⇒ Enantiomers have different biological properties in biological systems.

eg 1



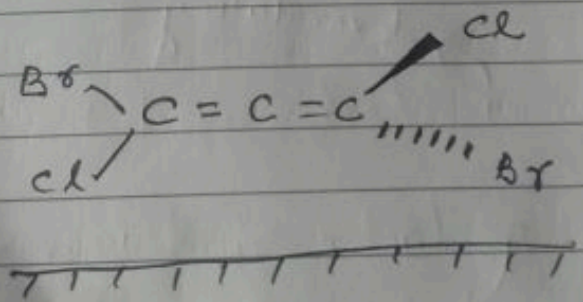
enantiomers

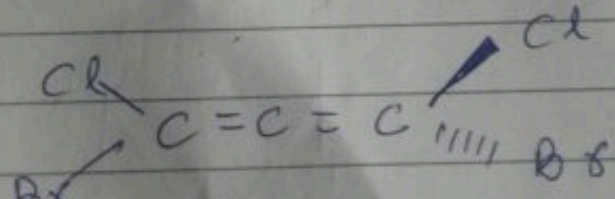
eg 2.



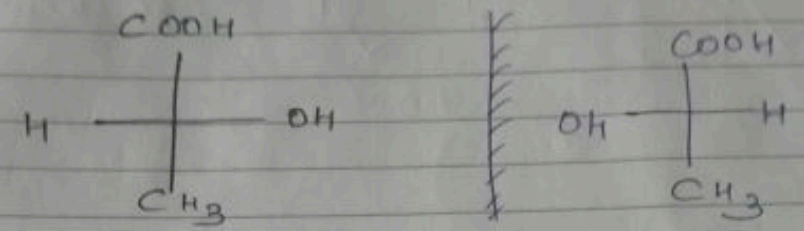
enantiomers

eg 3)





eg 4)



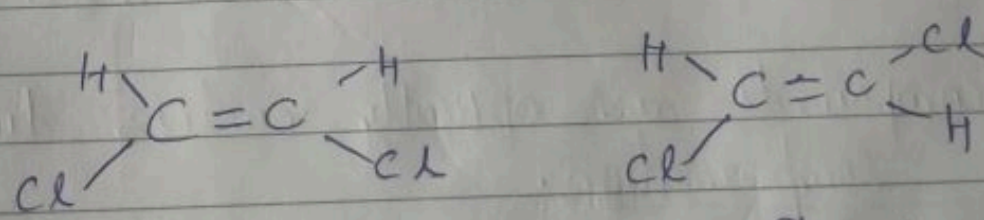
Lactic acid

[B] Diastereomers :- stereoisomers which ~~have~~ are not mirror image of each other and non-superimposable.

⇒ Diastereomers may be optically active or may not be optically active.

⇒ Diastereomers have different physical properties, so they can be separated by normal physical separation method.
(like fractional distillation, recrystallisation)

eg 1

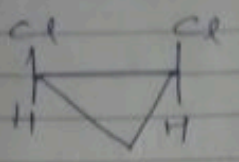


Cis

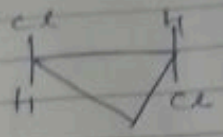
Trans

Diastereom

eg 2>

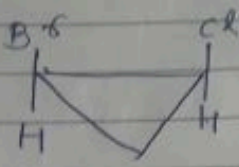


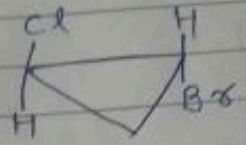
Cis



Trans (Diastereomers)

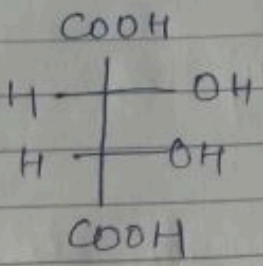
eg 3>



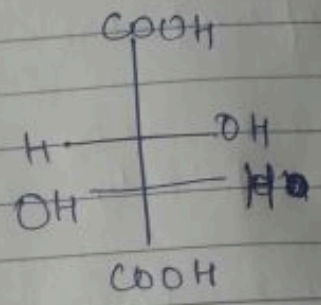


(Diastereomers)

eg 4>



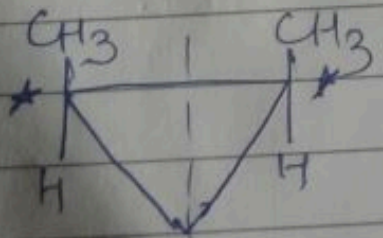
(V.V.I) (Tartaric acid)



(Diastereomers)

C> Meso compound :- chiral atom containing optically inactive compounds are meso compounds.

⇒ Meso compounds are optically inactive due to internal compensation.

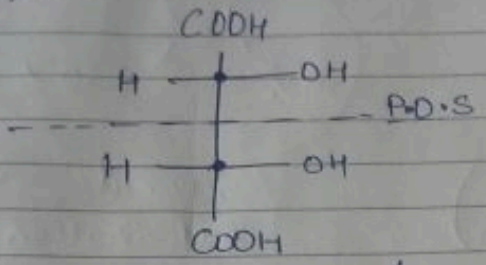


P.O.S = ✓

(achiral molecule) (optical inactive compound

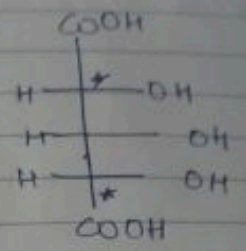
-> chiral atom present
⇒ Meso compound

eg 3>



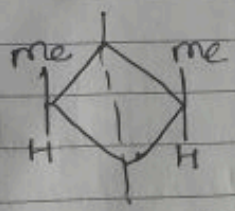
Meso tartaric acid

eg 3>



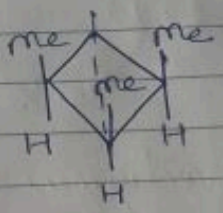
⇒ Meso compound
⇒ Optical inactive compound

eg 4>



⇒ P.O.S = ✓
⇒ Optical inactive
⇒ No chiral atom
⇒ Not a meso compound.

eg 5>

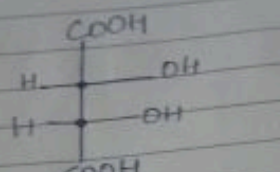


Meso compound

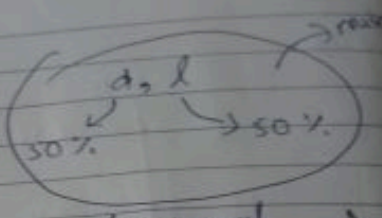
4> Racemic mixtuse = Racemic modification = (±) = dl

⇒ A mixture which contain equal amount of enantiomers.

⇒ A pure racemic mixture is optically inactive due to presence of external compensation.

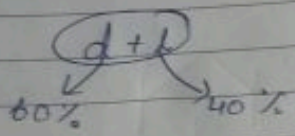


(internal compensation)



(external compensation)

⇒ Enantiomeric excess (ee) :-



when enantiomers are present in unequal amount in a mixture, then excess ~~from~~ percentage of predominant enantiomer is called enantiomeric excess. Its value is equal to value of optical purity for mixture.

$$e.e = \frac{|d-l|}{|d+l|} \times 100$$

and

$$\text{optical purity} = \frac{[\alpha]_{mix}}{[\alpha]_{pure}} \times 100$$

Q. Identify values of $[\alpha]$ mix for given compositions.

if $[\alpha]_{pure} = +80^\circ$

$[\alpha]_{mix} = \frac{ee \times [\alpha]_{pure}}{100}$

	% d	% l	ee	% Racemic mix	$[\alpha]$ mix
1)	50%	50%	0	100%	0°
2)	100%	0%	100%	0%	+80°
3)	60%	40%	20%	80%	+16°
4)	80%	20%	60%	40%	+48°
5)	25%	75%	50%	50%	-40°
6)	33.33%	66.67%	33.34%	66.66%	$-\left(80 \times \frac{33.33}{100}\right)^{\circ}$
7)	65%	35%	30%	70%	+24°
8)	30%	70%	40%	60%	-32°

E) Resolution :- we know enantiomers can't be seperated by normal physical separation method from a racemic mixture because enantiomers have identical physical properties.

·) Separation method of enantiomers from mixture of enantiomers is called resolution.

·) enantiomers are also known as resolvable compound.

Methods of Separation

Resolution

- Mechanical resolution
- Biological resolution
- Chromatography resolution.
- chemical method
  - Diastereomeric method.

Diastereomeric resolution (chemical method)

In this method racemic mixture is treated with pure optical active compounds and form mixture of diastereomeric products.

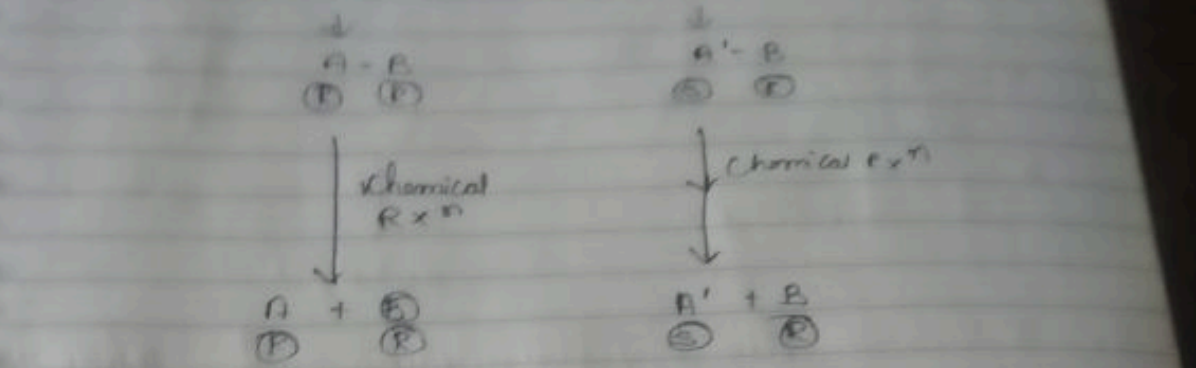
These diastereomeric products are isolable (by physical separation method). This isolated diastereomers may be regenerated and isolate pure enantiomers.

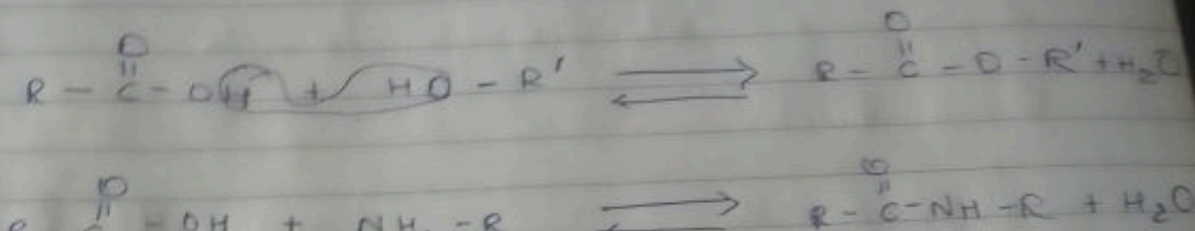
(R S / A + A') + B (R) pure optical active compound → [A-B + A'-B (S R / S R) mix.

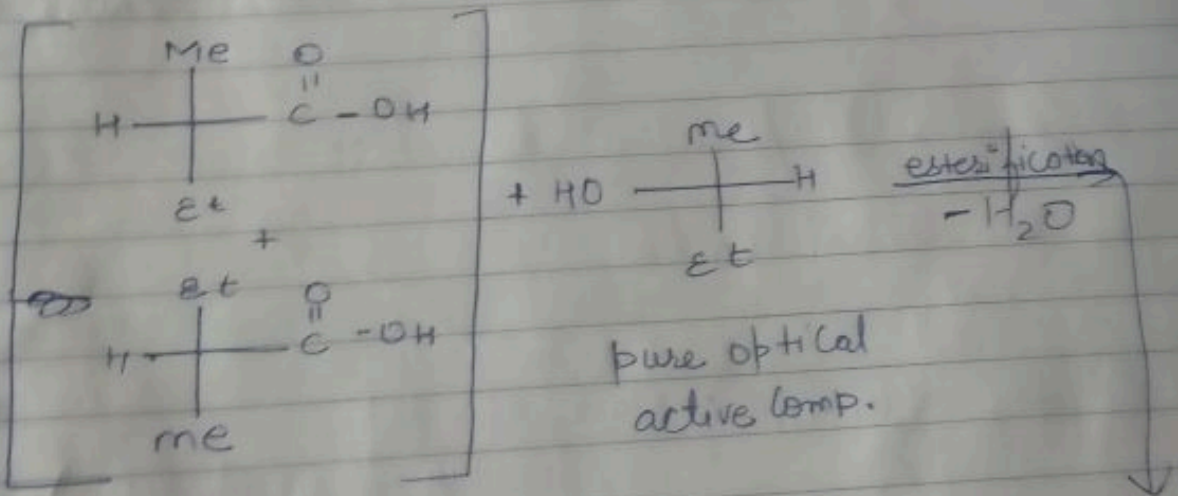
Racemic mixture

mixture of diastereomers (optical active) mix.

fractional distillation

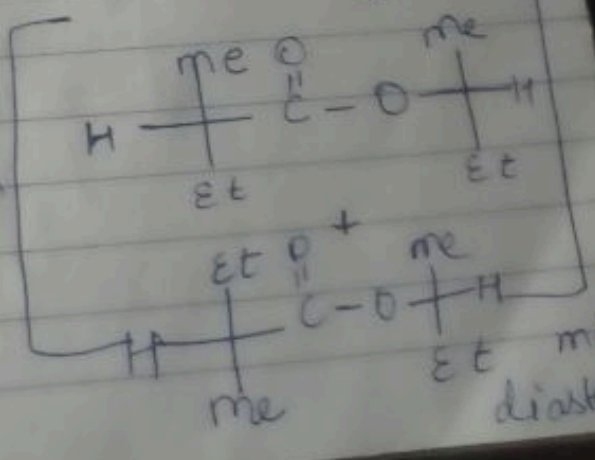


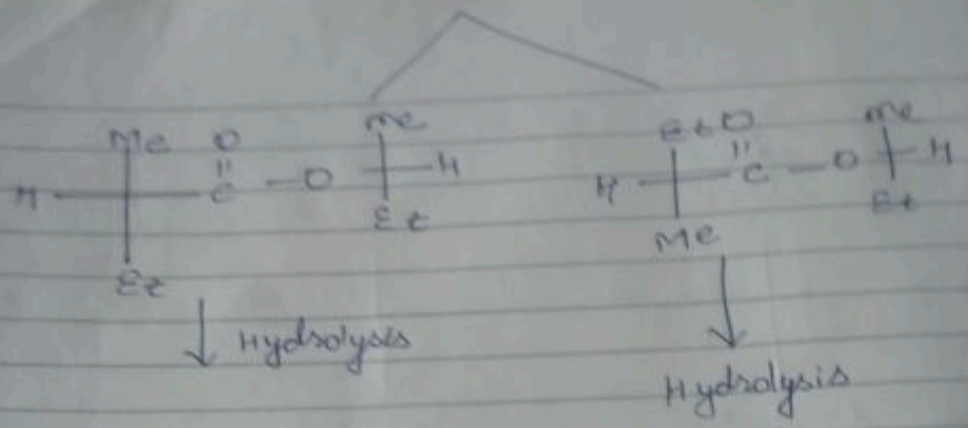




Racemic mix

fractional distillation (next page)



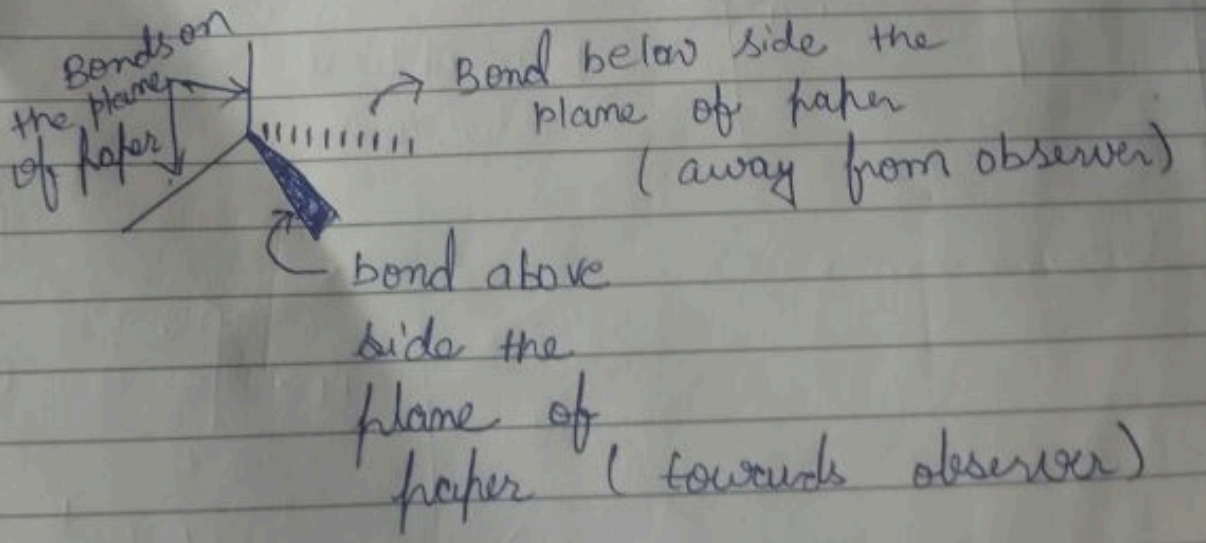


I) 3-Dimensional representation of molecules

A) → wedge dash projection formula
B) → Fischer " "
C) → Sawhorse " " } not applicable for only one chiral carbon.
D) → Newmann " "

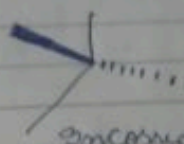
A) wedge dash projection formula:-

→ For one chiral carbon.

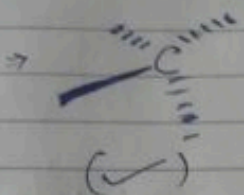


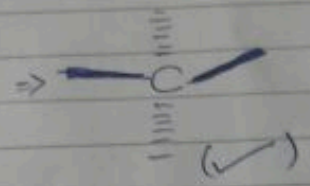


correct representation.

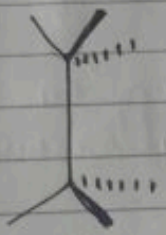


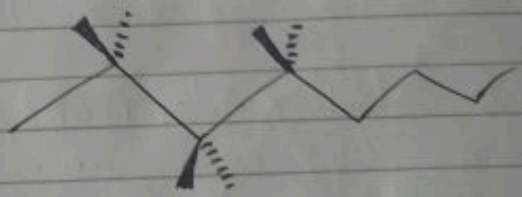
Incorrect representation



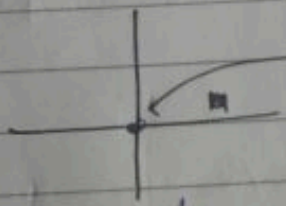


⇒ For more chiral carbons:

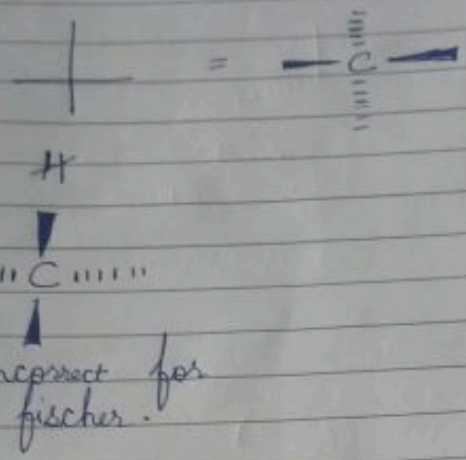




B) Fischer projection formula :-



- ⇒ Horizontal line represents bond above side of plane of paper. (towards observer)
- ⇒ vertical line represents bond below side of plane of paper. (away from observer)



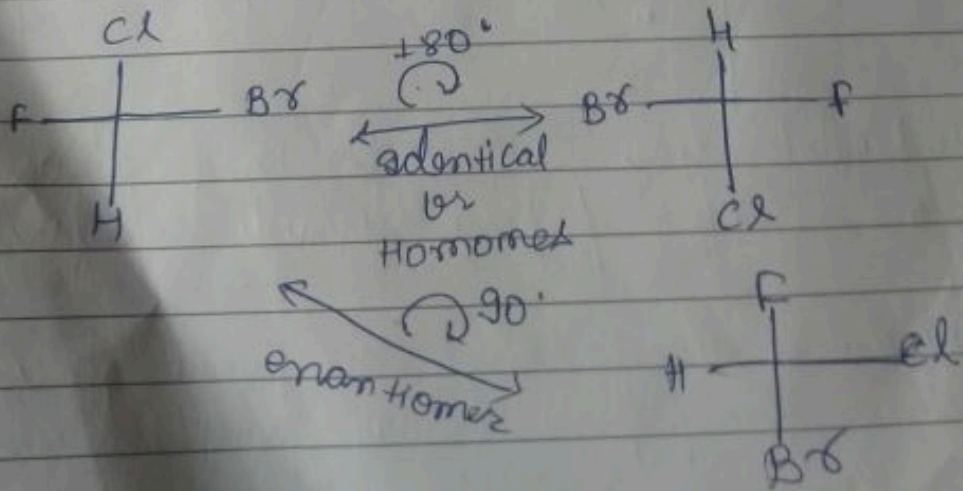
1. we obtain identical structure by 180° rotation of fischer formula in the plane of the paper.

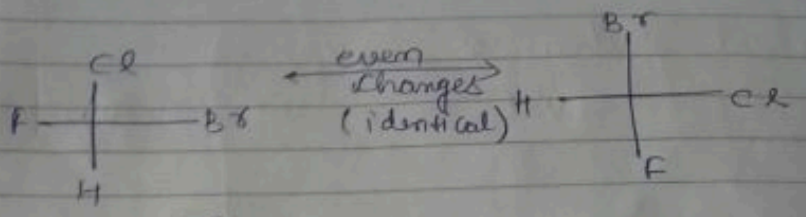
2. we obtain enantiomeric structure by 90° rotation of fischer formula in the plane of paper.

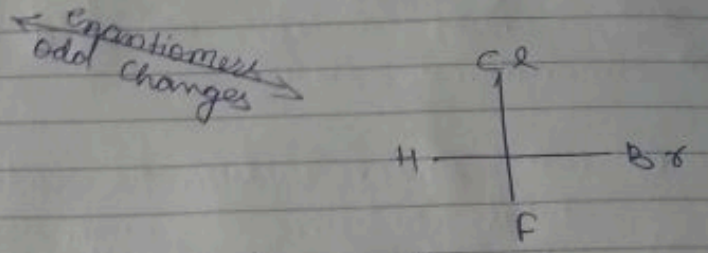
3. we obtain identical configuration on chiral carbon by even no. of simultaneous exchanges of ~~groups~~ group.

4. we get inversion in configuration of chiral ~~atom~~ carbon by odd no of simultaneous exchanges of groups on same chiral carbon.

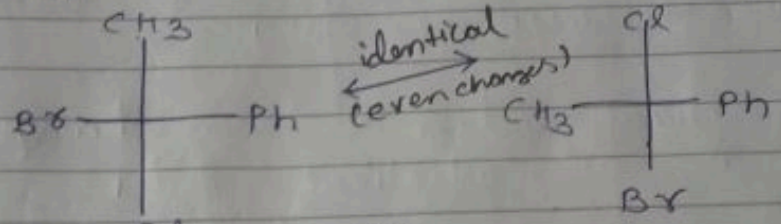
eg 1

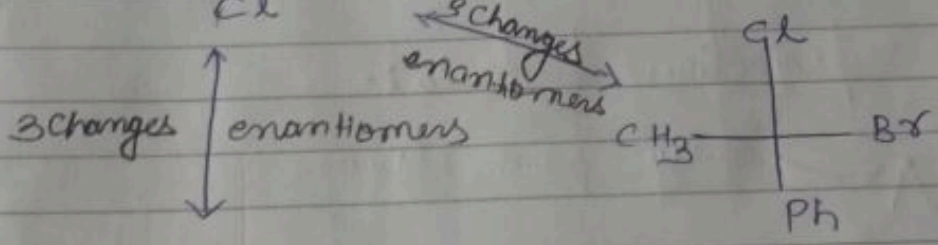


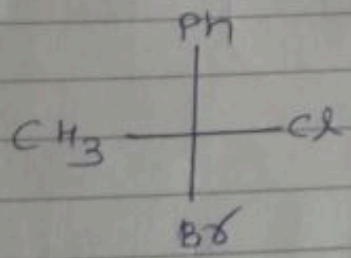

Cl
F
Br
H
even changes (identical)
Br
H
Cl
F

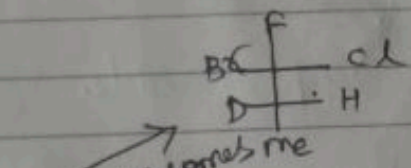

enantiomers
odd changes
Cl
H
Br
F

Q.

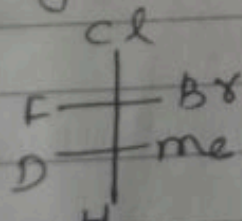

CH3
Br
Ph
Cl
identical
(even changes)
Cl
CH3
Ph
Br

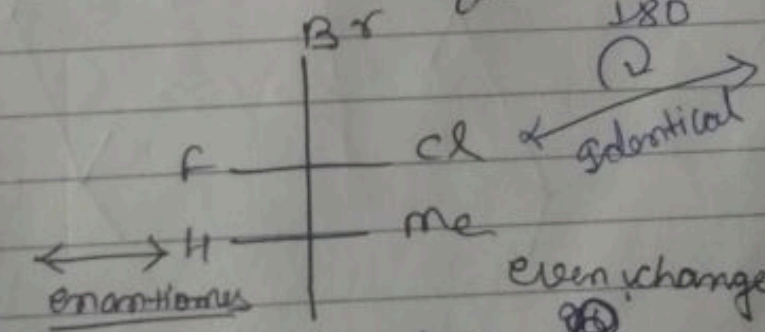

3 changes
enantiomers
3 changes
enantiomers
Cl
CH3
Br
Ph

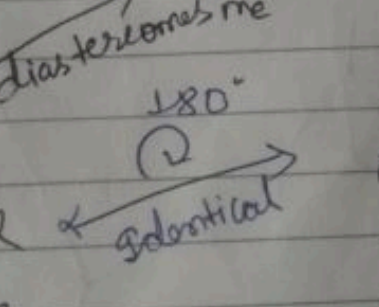

Ph
CH3
Cl
Br

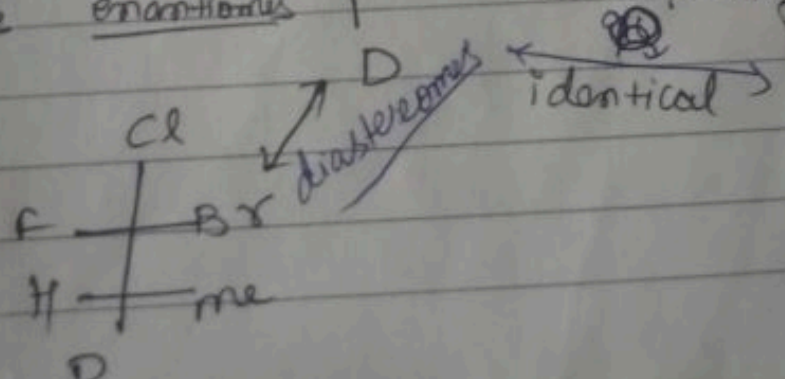

F
Br
Cl
D
H
diastereomers
me

eg 3)


Cl
F
Br
D
me
H


enantiomers
Br
F
Cl
me
H
D
180°
identical


D
me
H
Cl
F
Br


even change
identical
diastereomers
Cl
F
Br
H
me
D
F
Cl
Br
me
D
H

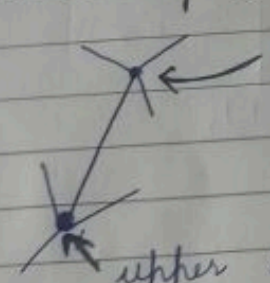
on 2 chiral carbon or more
हमलोग देखा chiral carbon को आपस - उपस
treat करेंगे

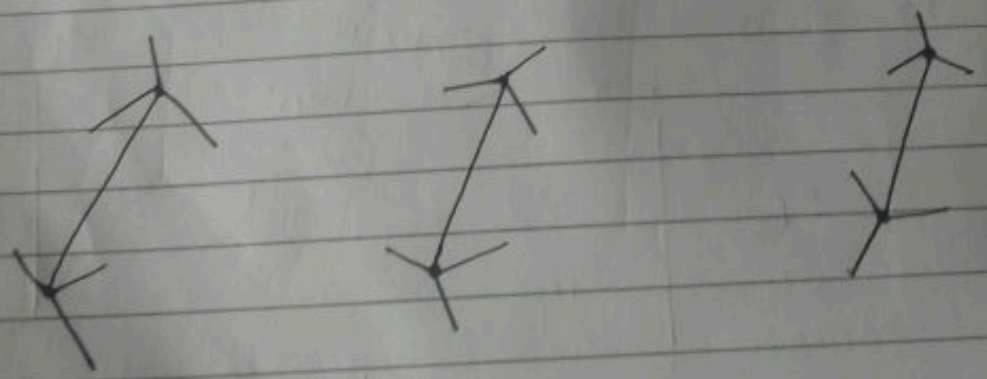
-> अगर दोनों chiral carbon पर odd change होगा
तो वो enantiomer होगा

-> अगर एक chiral carbon पर odd change
और दूसरे chiral carbon पर even change / NO change
होगा तो वो diastereomers होंगे।

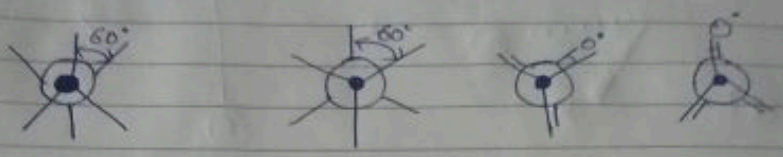
-> अगर दोनों chiral carbon पर even change / NO-
change होगा तो वो identical होंगे।

c> Sawhorse projection formula :-



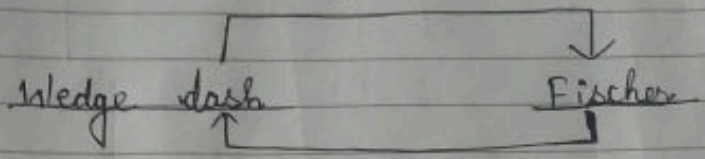


D) Newmann's projection formula:

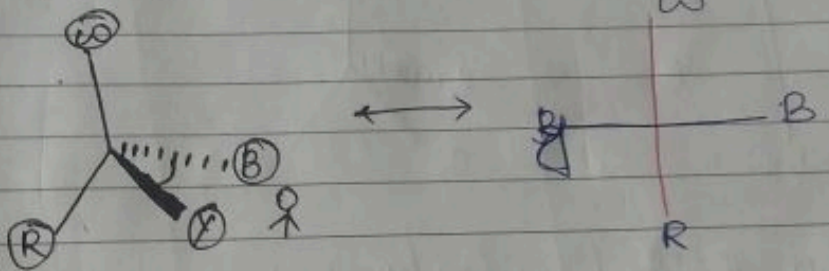


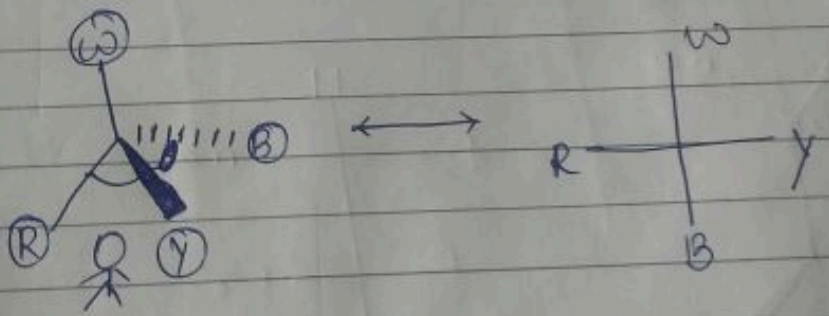
X) Interconversions of 3D projection formula

A)



1) For one chiral carbon :-

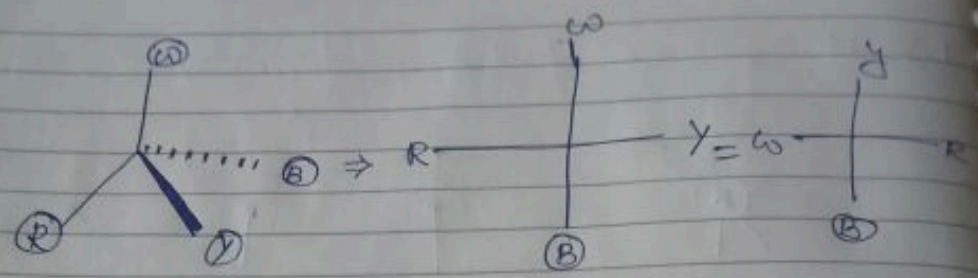


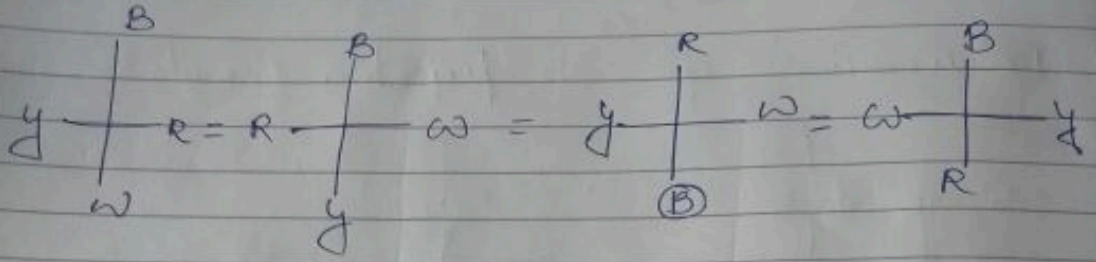


Short trick

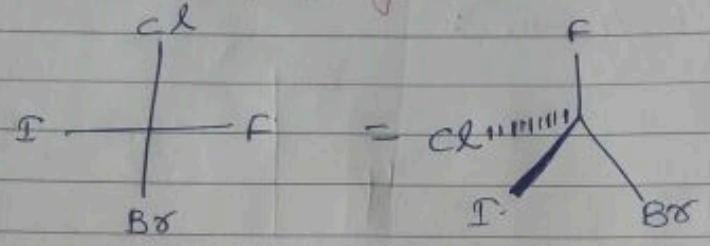
* placedotted group on vertical lines

⇒ Remaining groups in same sequence on remaining position.

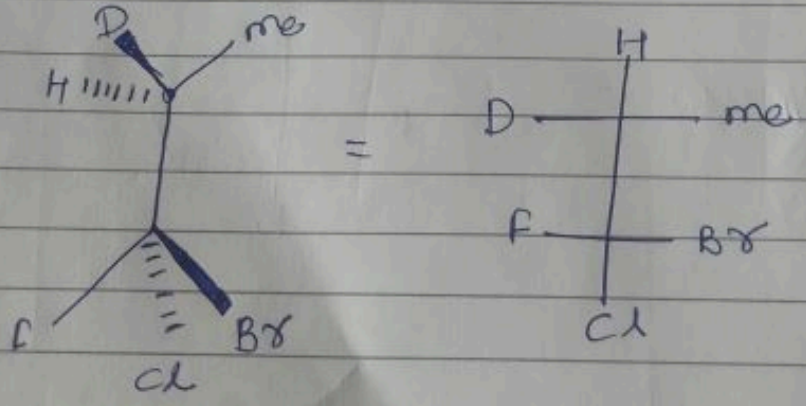




Fischer → wedge dash



For two chiral carbons :

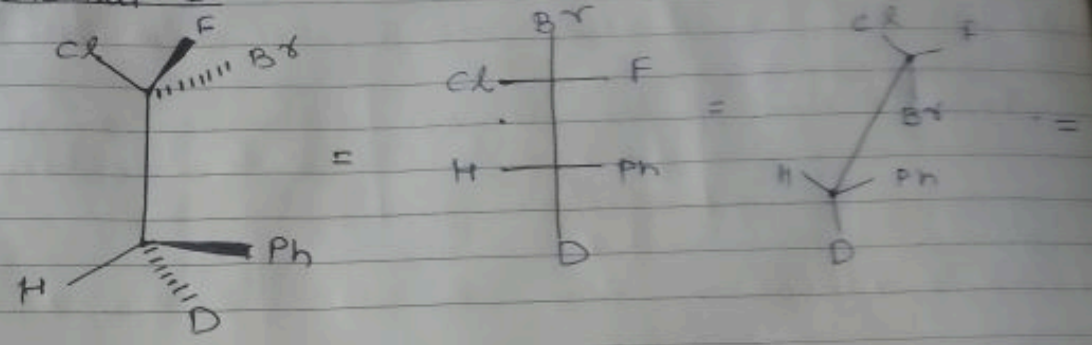


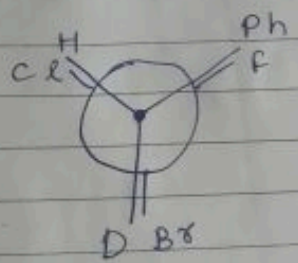
⇒ जो group ≡ bond पर है उसको vertical line पर place करना है फिर जो right में है उसको right में और जो left में है उसको left में।

Q)

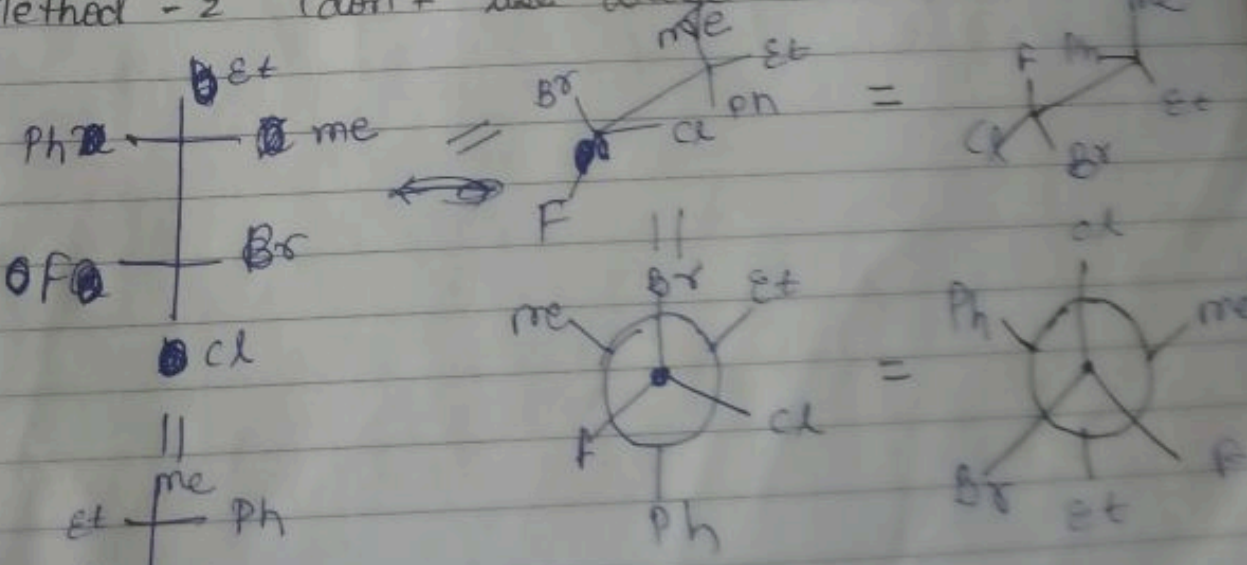
B) For wedge dash, Fischer, Newmann's and Sawhorse.

Method 1





Method - 2 (don't use wedge dash)

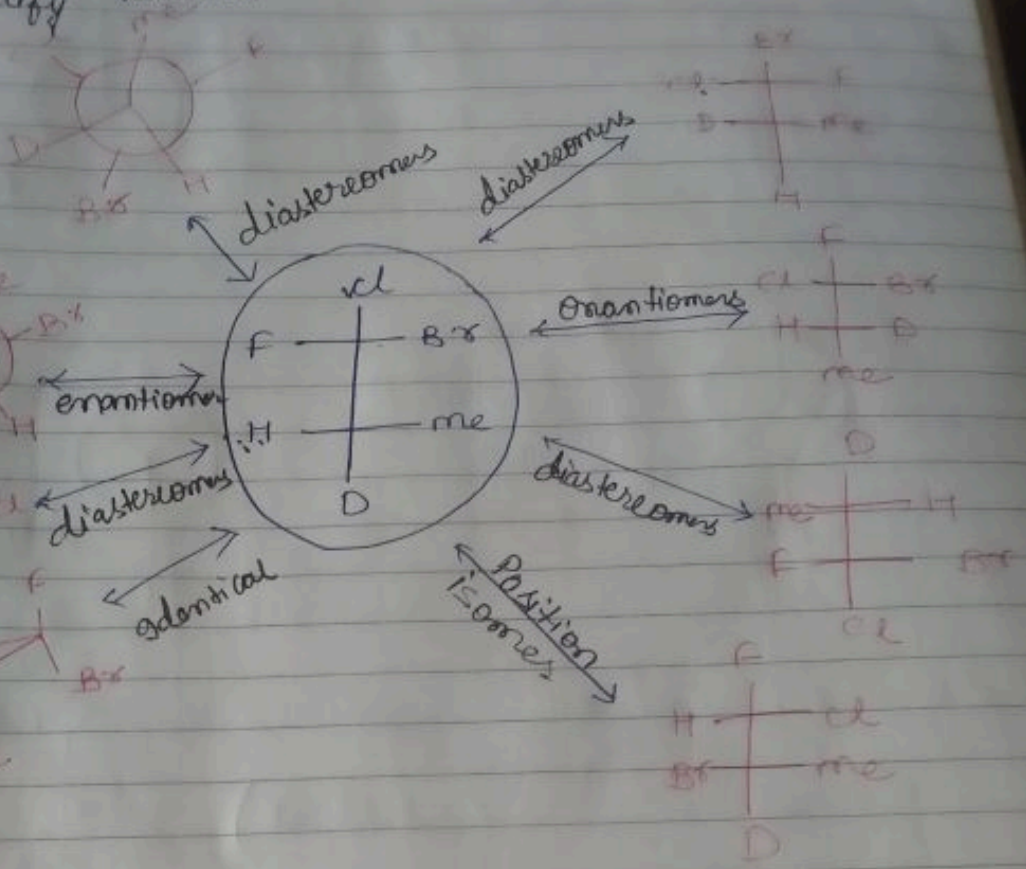


Method 2 में जो Fischer के नीचे लगा chiral Carbon है उसको उसके नीचे है उसी Sense में Neumann and Sawhorse में रखना है पर जो Fischer के ऊपर वाला chiral Carbon का group है उसको उसके ऊपर के उसका उल्टा sense में रखना है।

For Method - 2 (Chart, don't use wedge dash)

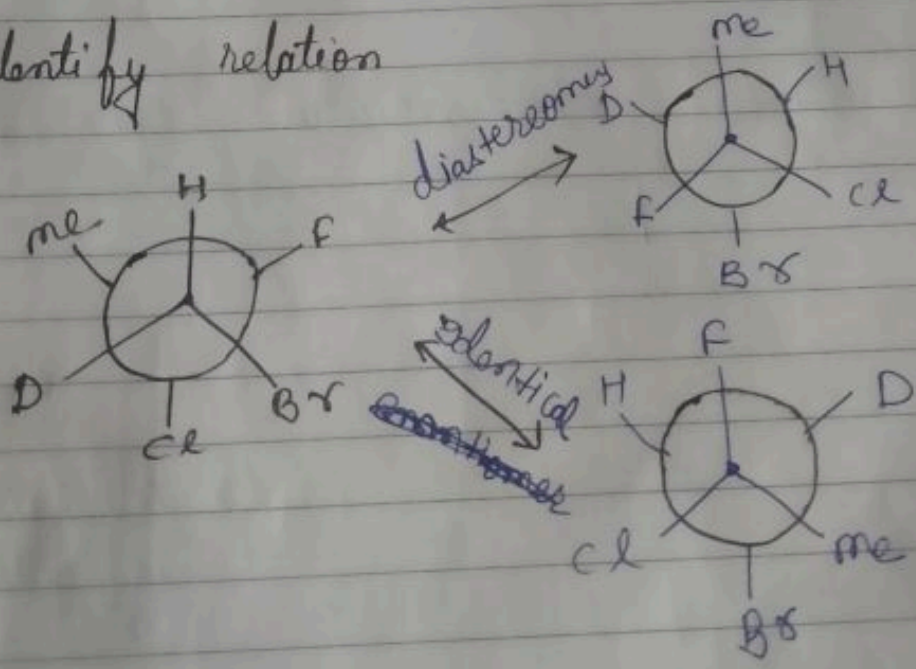
Interconversion	Sequence on front side Carbon	Sequence on back side Carbon
1) Fischer ↔ Fischer	Same Sequence	Same Sequence
2) Fischer ↔ Sawhorse	" "	Opposite "
3) Fischer ↔ Neumann	" "	Opposite "
4) Sawhorse ↔ Sawhorse	" "	Same "
5) Neumann ↔ Neumann	" "	Same "
6) Sawhorse ↔ Neumann	" "	Same "

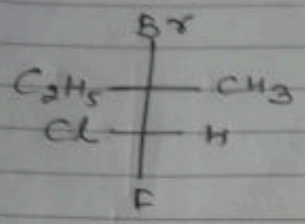
Identify Relation



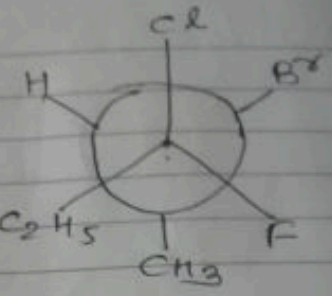
Identify relation

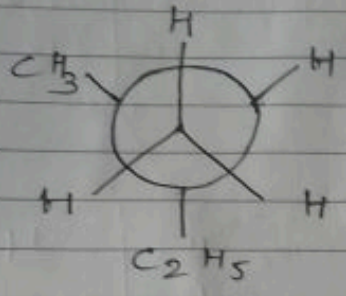
1)



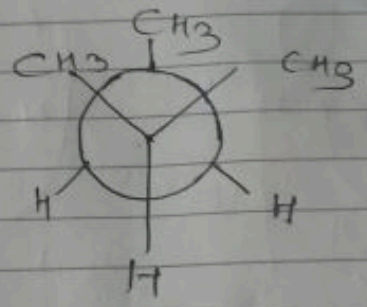
2) 

Chain ~~Positional~~ ~~Isomer~~ Isomer



3) 

positional Isomer



$CH_3 - CH(CH_3) - CH_2 - CH_3$ (Identical) $CH_3 - CH(CH_3) - CH(CH_3)_2$

XI) Configurational Nomenclature of chiral carbons

A) Relative Configuration

i) → D-L Nomenclature

ii) → Threo-Erythro nomenclature

B) Absolute Configuration

→ R-S Nomenclature

A) i) D-L Nomenclature

This nomenclature is applicable for carbohydrates, amino acids and there derivatives.

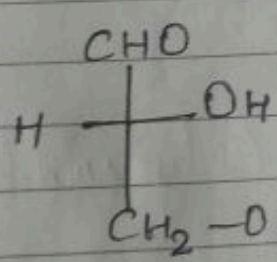
⇒ Carbohydrates :- Poly hydroxy carbonyl compounds.

⇒ Amino acids :- α-Amino Carboxylic acid.

⇒ For identification of D-L Nomenclature draw a fischer ~~three~~ formula of given compound, in which first carbon (most oxidised carbon) present on top of vertical line and last carbon at on bottom of vertical line. then compare with reference compound.

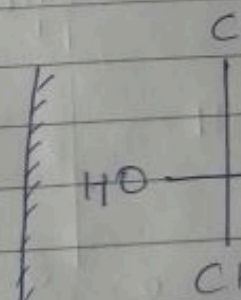
Reference Compounds

For carbohydrate



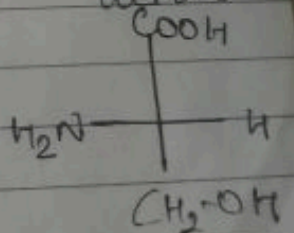
D(+)- Glyceraldehyde

Reference



L-(-)-Glyceraldehyde

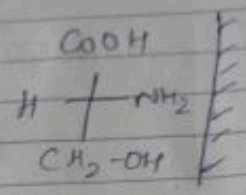
For amino acids



L-(+)-(Serine)

Reference

#Raj9705



D(+) serine

For carbohydrate :-

When -OH group of highest numbered chiral carbon is present on R.H.S then configuration will = D.

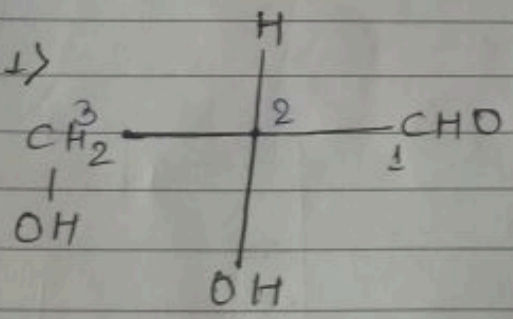
L.HS -- L

For amino acids

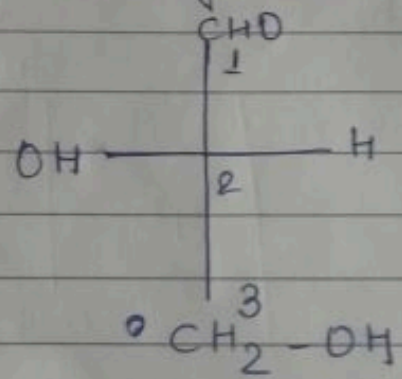
When NH_2 group of α position of -COOH group present on R.H.S = D

L.H.S = L

eg 1)



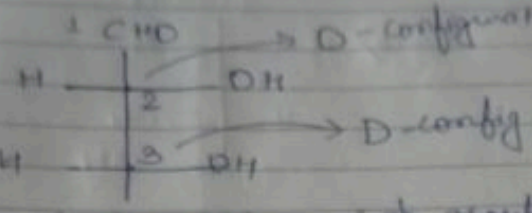
According to rule



(Glyceraldehyde)

L series

eg 2)



D-configuration

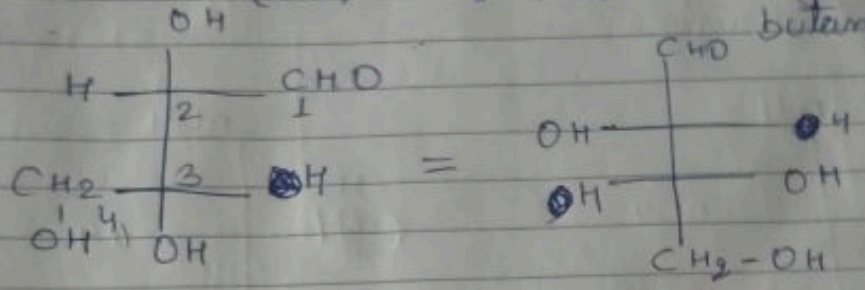
D - series

D - Erythrose

(compound D group वाला है bcz carbon D group वाला है)

(2D, 3D) - 2, 3, 4 - trihydroxy butanal

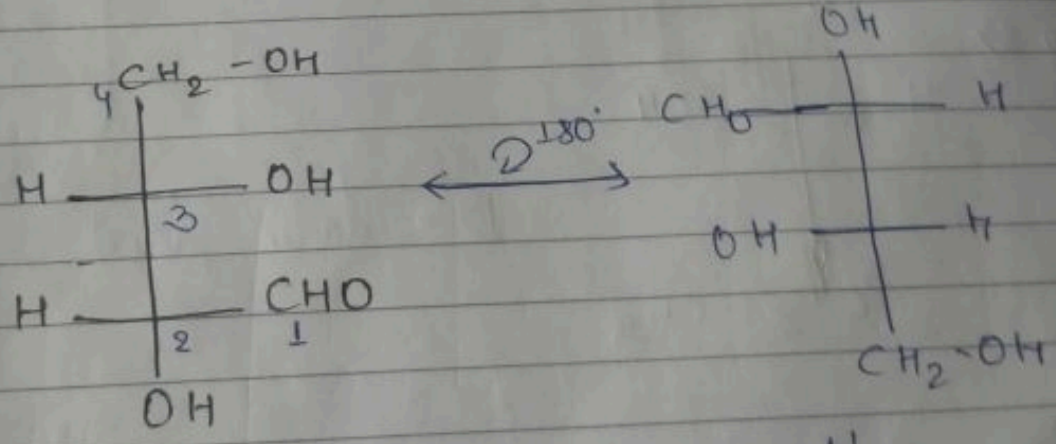
eg 3)



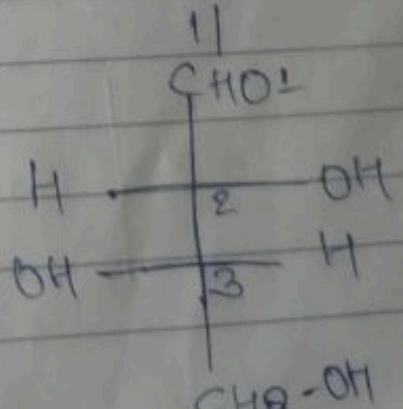
(D - series)

(D - Threose)

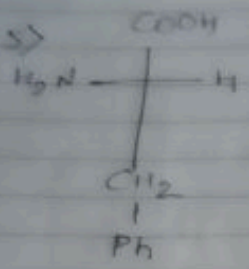
eg 4)



(L series)



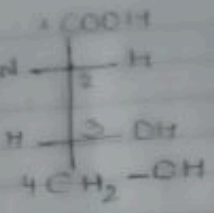
eg 5)



L - Series (amino acids)

L - Phenyl alanine

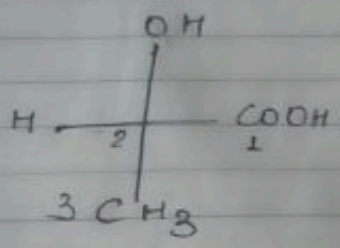
eg 6)



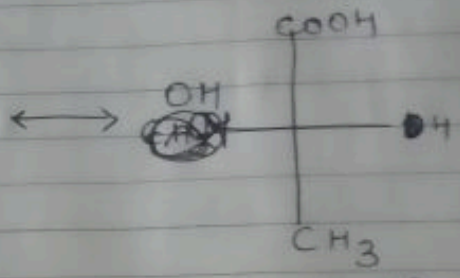
(L - Amino acid)

(L - Threonine)

eg 7)

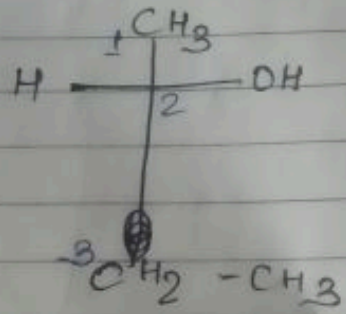


(Lactic acid)



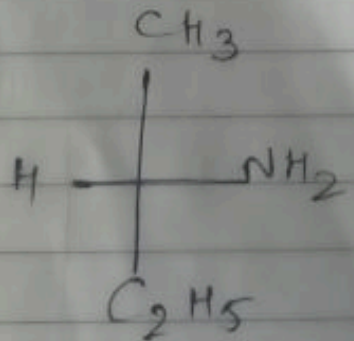
L - lactic acid

eg 8)



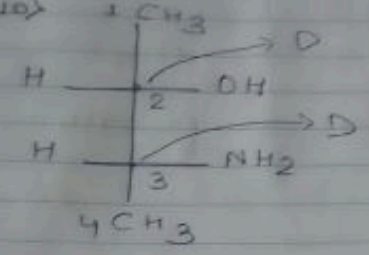
✗ - ✗tic ✗d

eg 9)

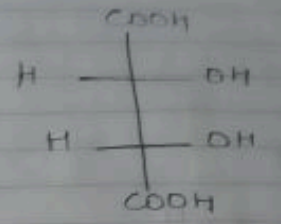


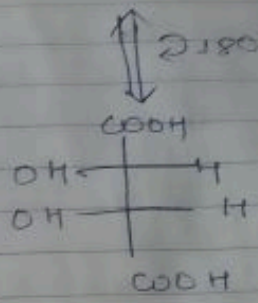
(D - amine)

eg 10)



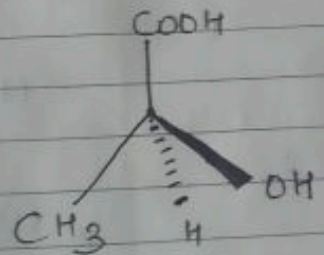
eg 11)



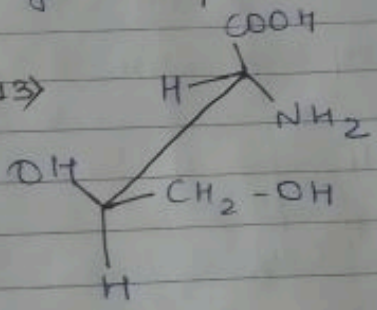


Note: D - L Nomenclature is not applicable for dissimilar ends containing compounds.

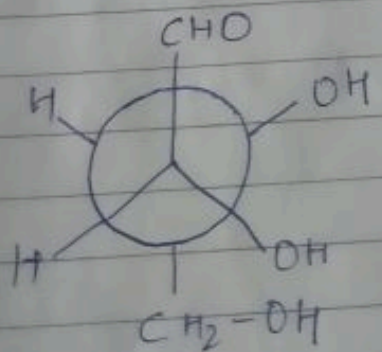
eg 12)



eg 13)

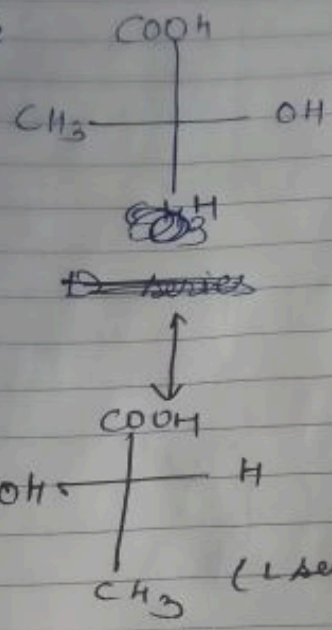


eg 14)



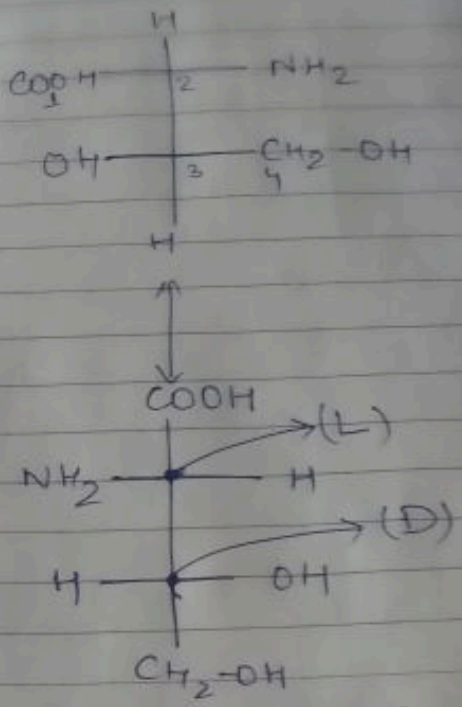
Ans

(a)



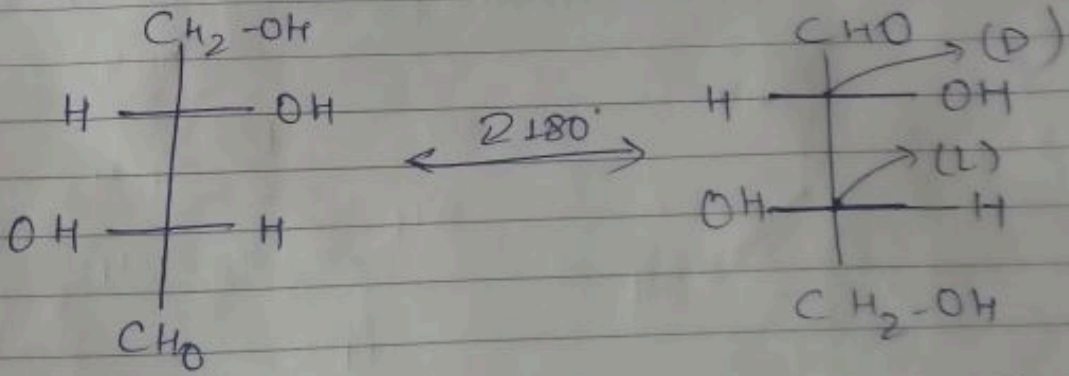
L-lactic acid

(b)



L-amino acid

(c)



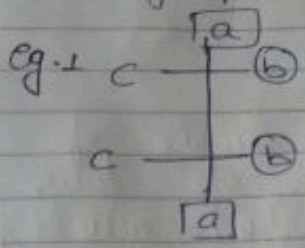
(L series)

L-carbohydrate

A·2> Threo - Erythro Nomenclature

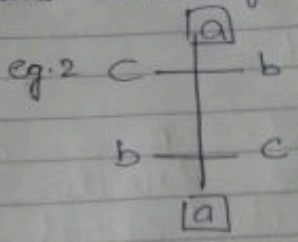
this Nomenclature is applicable for two chiral carbon containing compounds in which at least 2-2 groups should same of both chiral carbon.

eg·1



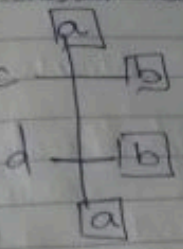
Erythro

eg·2



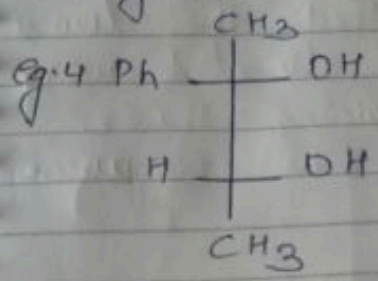
Threo

eg·3



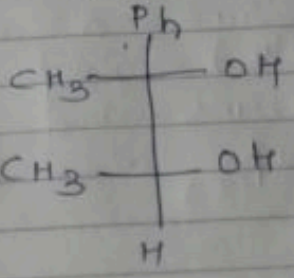
Erythro

eg·4



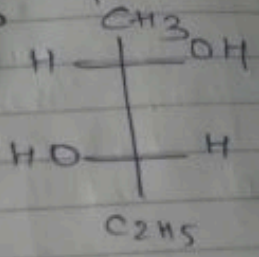
Erythro

eg·5



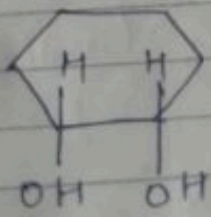
Erythro

eg·6



Threo

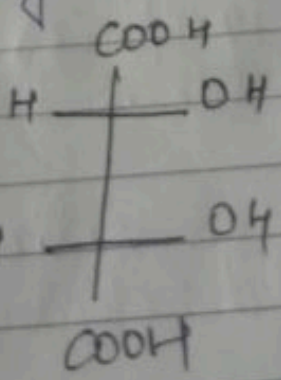
eg·7



Erythro

Notes: Meso compounds always be Erythro.

eg·8



B) Absolute Configuration

R-S Nomenclature

R = Rectus = Right

S = Sinister = left

For identification of R-S nomenclature of chiral carbon • give priority order of four different groups of chiral carbon according to "CIP sequence rule" and visualise the molecule • in such a way that the lowest priority group (4th) group is directed away from the observer. and remaining three (1, 2, 3) groups are facing to observer.

if Rotation of 1 → 2 → 3 group ↻ (Clockwise rotation) = R

" " " " " group ↺ (anticlockwise ") = S

⇒ Draw fischer projection formula of given compound.

⇒ Give priority order of four groups of chiral carbon according to CIP sequence rule.

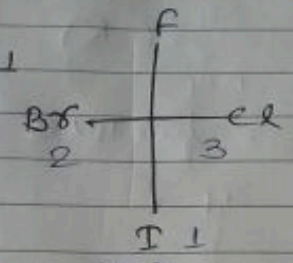
⇒ when 4th group is present on vertical line and Rotation of 1 → 2 → 3 ↻ (clockwise Rotation) = R

$-C=O$ $-\overset{O}{\overset{\|}{C}}-O-C$

↻ and $1 \to 2 \to 3$ ↺ (anticlockwise rotation) ⇒ S

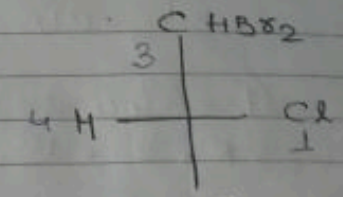
⇒ when 4th priority group on horizontal line and $1 \to 2 \to 3$ ↻ = S or $1 \to 2 \to 3$ ↺ = R

eg. 1



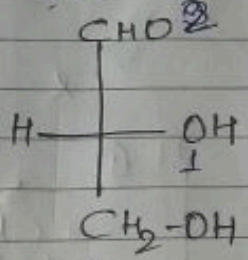
(R)

eg. 2



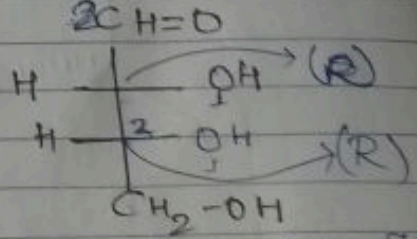
2CI3 (S)

eg. 3

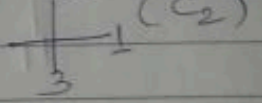


(R)

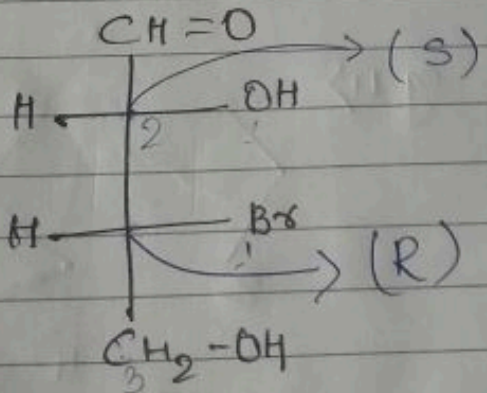
eg. 4



(2R, 3R) 2,3,4-Trihydroxybutanal

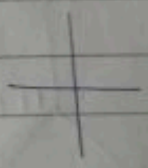


eg. 5

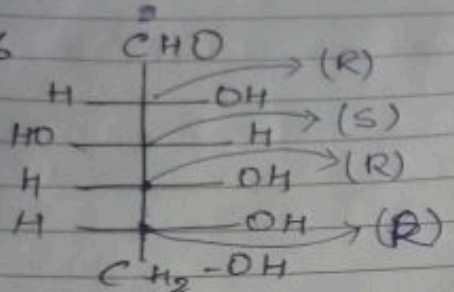


(2S, 3R)

(C3)



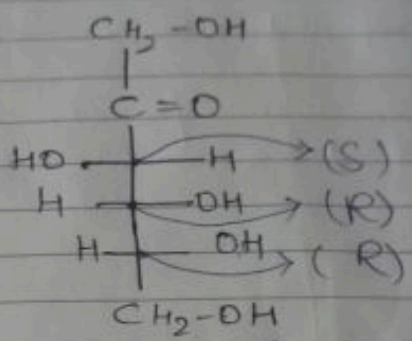
eg. 6



(D-Glucose)

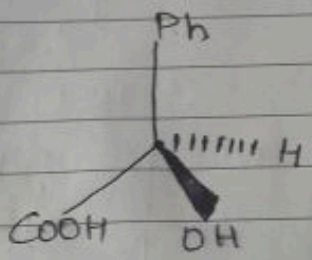
2R, 3S, 4R, 5R

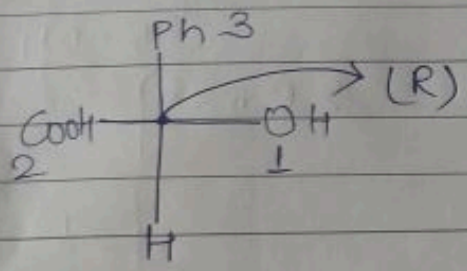
eg 7



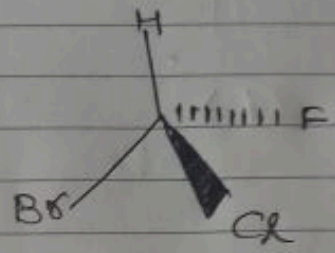
D- fructose

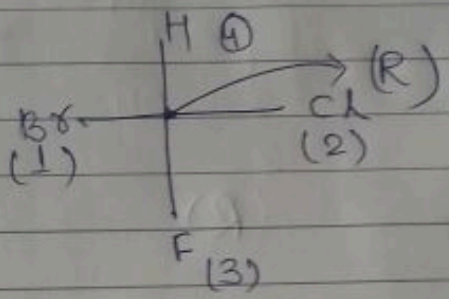
eg 8)



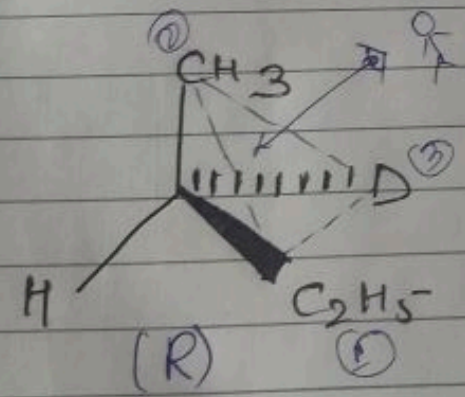


eg 9)

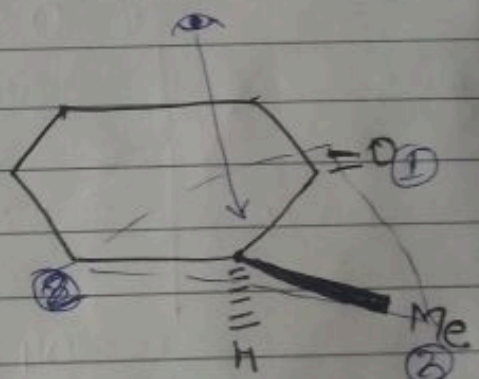




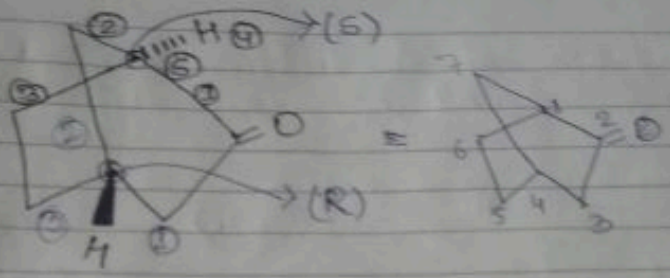
eg 10)



eg 11)

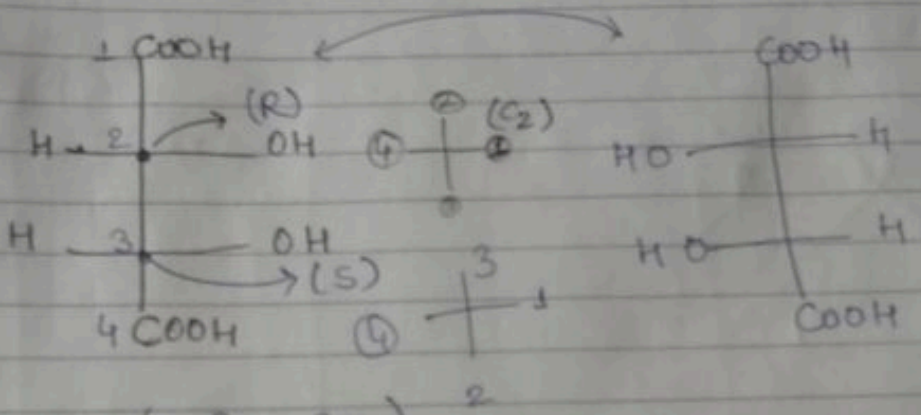


Eg 12)



(1S, 4R) Bicyclo (2,2,1) hepta - 2 - one

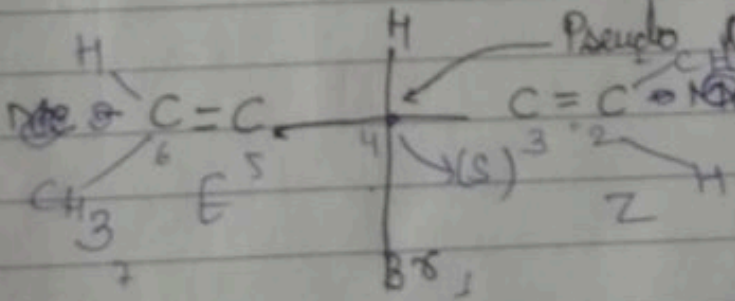
Eg 13)



(2R, 3S)

R > S R is more importance than S

Eg 14)



Pseudo chiral carbon

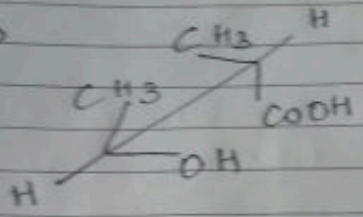
Cis > trans

Z > E

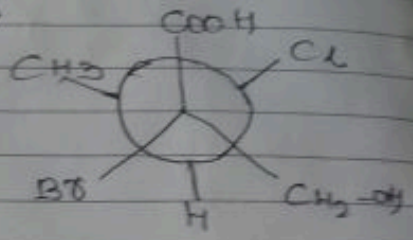
(2Z, 4S, 5E) 4-bromo hepta (2,5) d

Note : For pseudo chiral carbon, we use small d and small s for configuration.

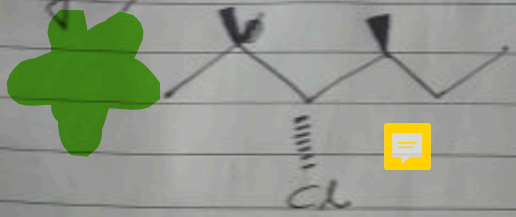
eg 15)



eg 16)

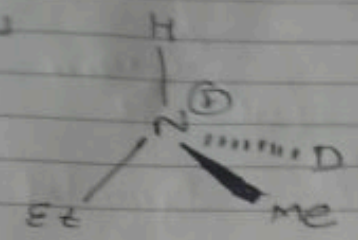


eg 17)



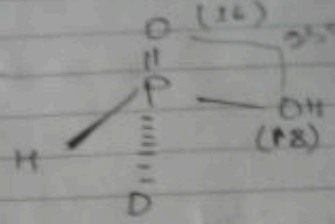
Examples of chiral centres (other than Carbon)

eg.1



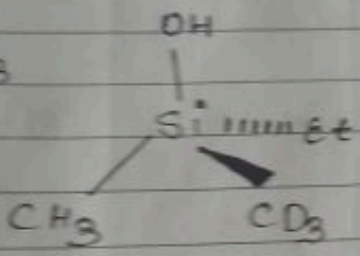
POS = X
COS = X
Tetrahedral (T_d)
(optically active compound)

eg.2



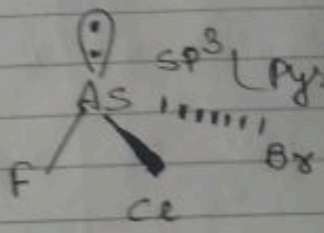
POS = X
COS = X
Tetrahedral (T_d)
(O.A.C)

eg.3



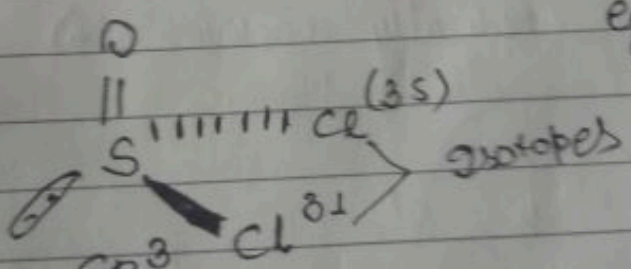
POS = X
COS = X
(O.A.C)

eg.4



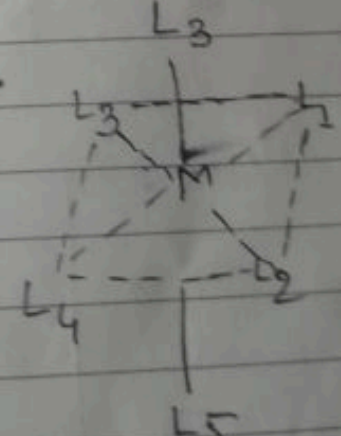
POS = X
COS = X
(O.A.C)

eg.5



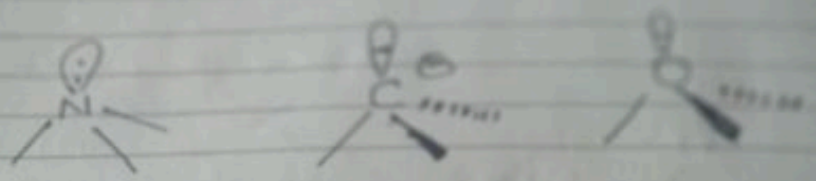
POS = X
COS = X
} O.A.C

eg.6



(OC + a Pedral)

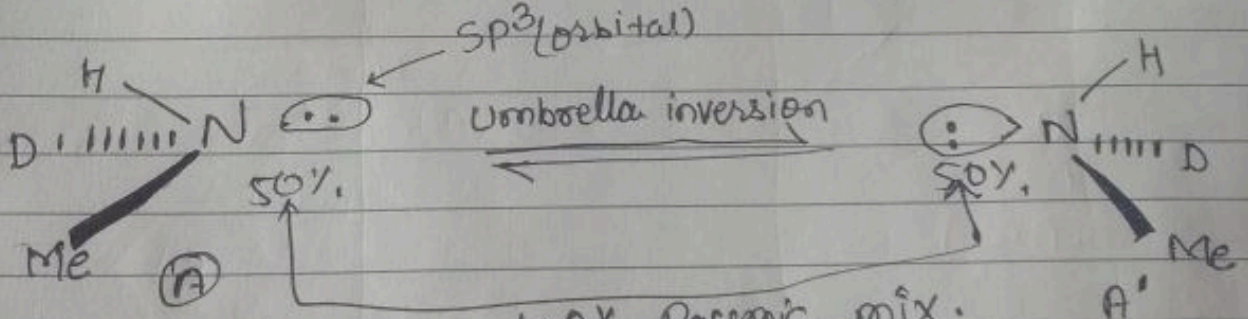
11. Umbrella inversion :-



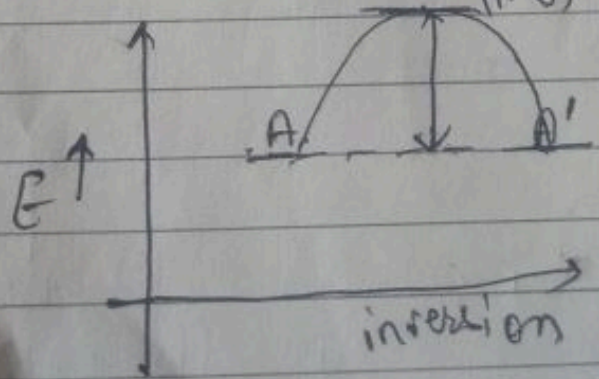
Pyramidal Pyramidal Pyramidal

Note: When N, C and oxygen have three different groups with one lone pair then this type of structures are optically inactive (at room temp). It is due to umbrella inversion. By is inversion above species form racemic mix.

Examples of chiral centres (other than Carbon)



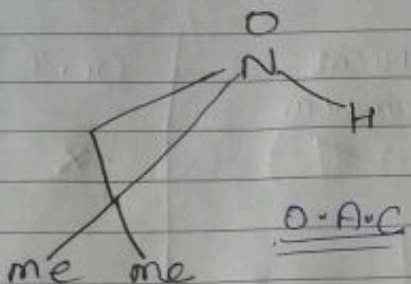
(T.S) Planar



2×10^{13} rps

Note: Umbrella inversion not possible for 2nd and higher period elements at room temp.

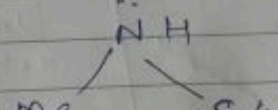
→ When asymmetric Nitrogen is present in small ring (upto 5 membered ring) then umbrella inversion not possible due to high angle strain in intermediate ~~structure~~ (T.S). Such type of compound are optically active.



O.A.C

(Racemic mixture)

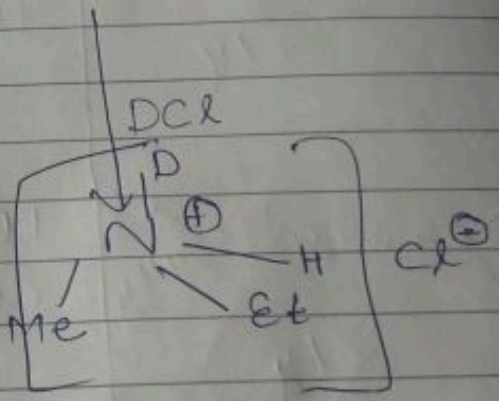
eg.



2° amine

(Optically inactive mixtures)

Non-isolable.



(Racemic mixture)

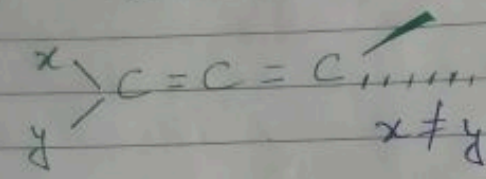
(isolable)

(Umbrella inversion not possible)

Examples of optical active compounds Without chiral atom

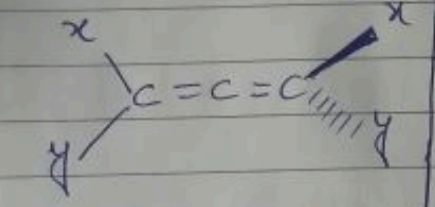
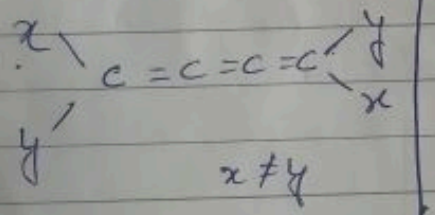
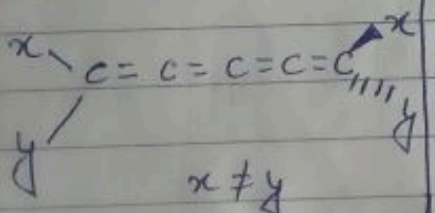
[A] Cumulenes = 0

Eg:



(Allenes)

~~Terminal xy groups~~

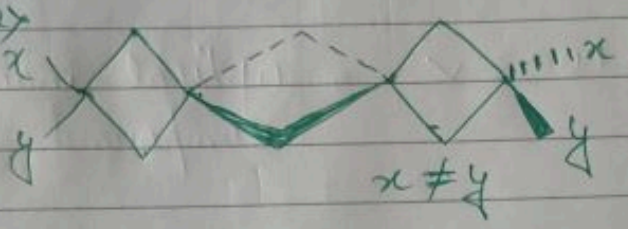
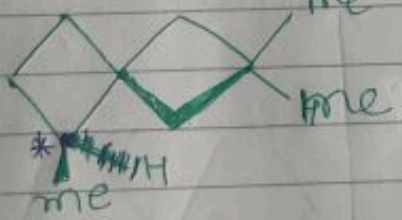
	Terminal group x, y groups	Optical isomerism	Cr. I
	⊥ POS = x COS = x	✓	x
	in same plane POS = ✓	x	✓
	POS = x COS = x	✓	x

Condition for optical activity in cumulenes

i) Terminals $x \neq y$

ii) even no. of C=C double bond

B) Spiro compounds

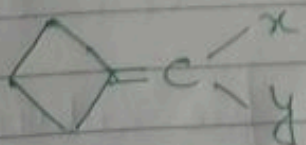
eg ⇒ 	$x \neq y$	POS = ✓ COS = ✓	Optical isomerism (✓)	G.I = (✓)
eg 2) 	$x \neq y$	POS = ✓	O.I = ✗	G.I = ✓
eg 3) 		optical active due to one chiral atom		

conditions for optical activity in Spiro compounds

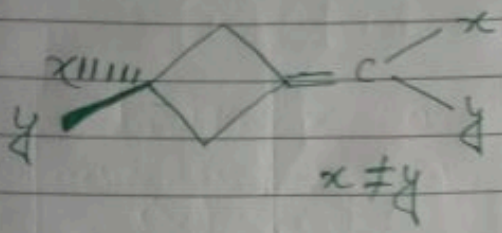
i) $x \neq y$
ii) even no. of Spiro ring.

(C) Alkylidene cycloalkene :-

$CH_2 - CH <$ (Alkylidene)

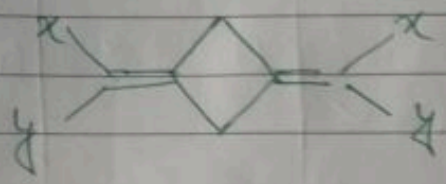


eg 1)



POS = X
COS = X , optical active Comp.

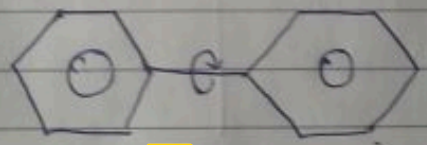
eg 2)

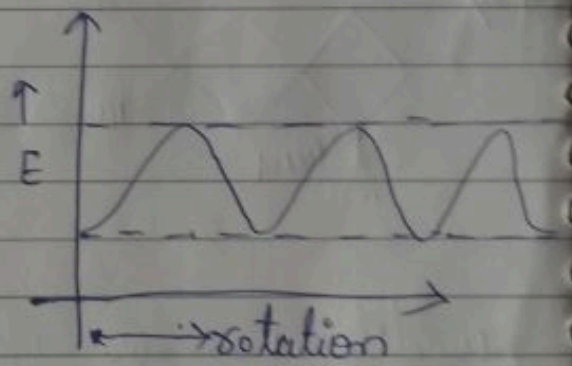


POS = ✓ , optical inactive compound.

ORTHO substituted biphenyl (Ph-Ph)

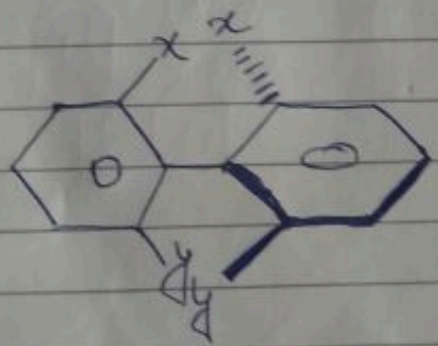
eg 1)





(Atrop isomerism)

eg 2)



Condⁿ:- $r_x + r_y \geq 2.9\ A^0$

(if this Condⁿ is possible then group दूर हो जाते हैं और same plane में नहीं रहते हैं।)

⇒ POS = X
⇒ COS = X

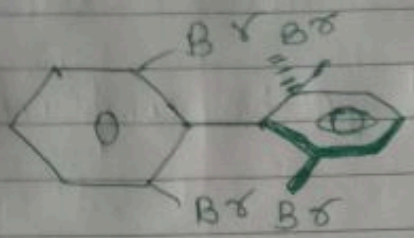
optical active

(Restricted rotation) (No free rotation)

⇒ x ≠ y ≠ H ≠ D ≠ F (etc)

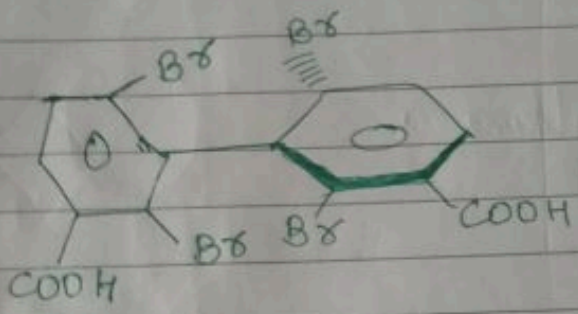
Physics Chemistry

eg 3)



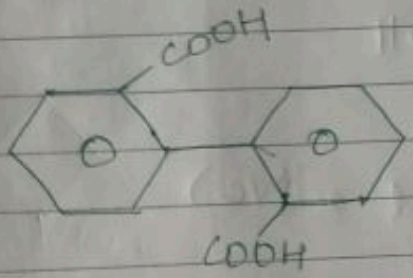
No free rotation
⇒ POS = ✓
⇒ optical inactive

eg 4)



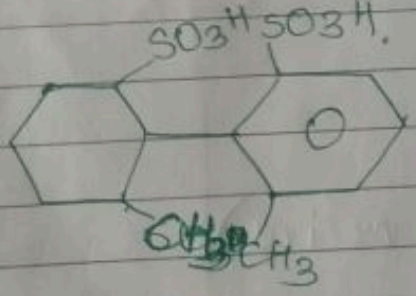
⇒ POS = X
COS = X
= optical active

eg 5)



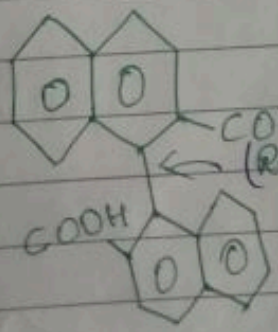
(optical inactive)

eg 6)



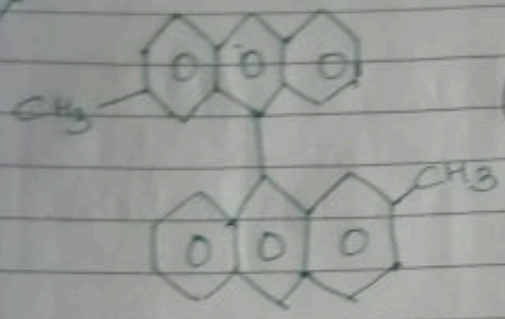
(optical active)

eg 7)



(Restricted rotation)

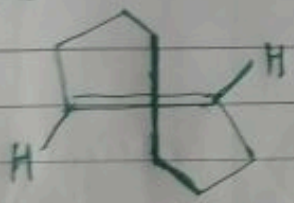
eg 8)



(optical active compound)

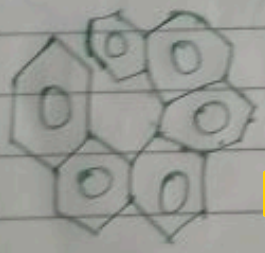
(E) Trans cyclo alkene :-

eg 1) Trans cyclo octene



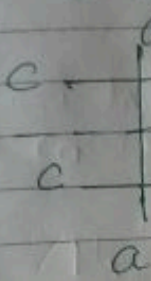
(optical active)

Other example

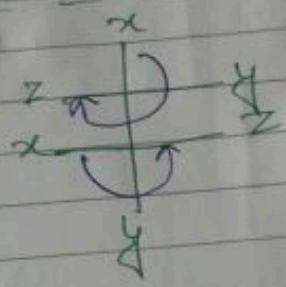


(optical active)

Short trick for identification of meso compound

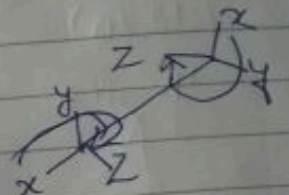


(Meso)

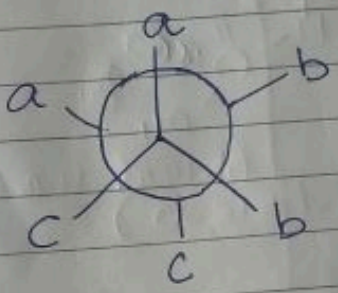


(Meso)

for Sawhorse and Newmann



(Meso)

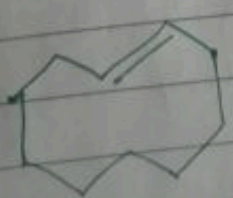


(meso)

NOTES: meso compound पहचानने के लिए fischer projection में एक clockwise और एक Anticlockwise होना चाहिए उसकी Newmann and Sawhorse में दोनों का rotation same होना चाहिए।

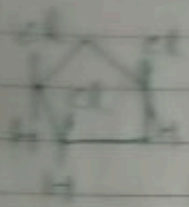
Q Identify optical active, optical inactive and meso compounds :-

Q. 1)



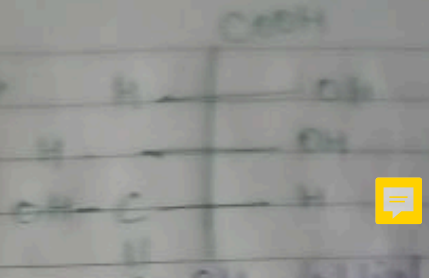
(optical inactive)

2.



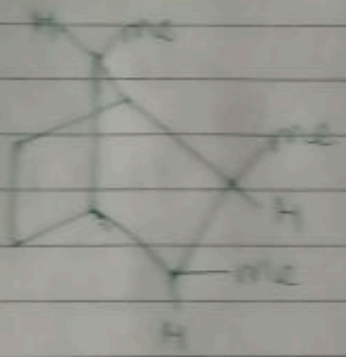
(Optical inactive)
(Meso)

3)



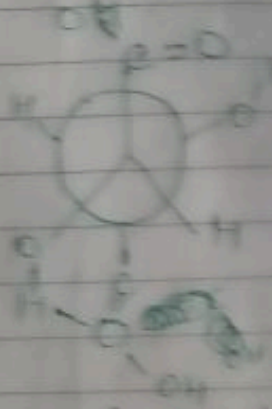
(Optical inactive)
(Meso)

4)



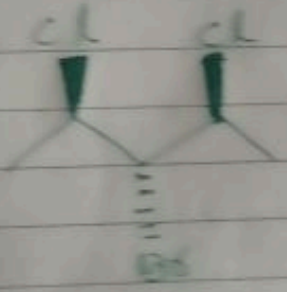
(Optical inactive)
(Meso)

5)



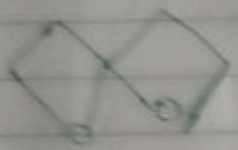
Optical inactive
(Meso)

6)



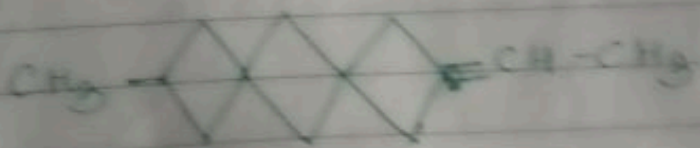
(Meso)
(Optical inactive)

7)



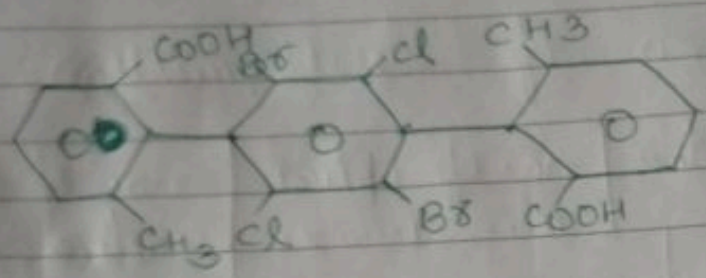
(Optical active)

8)



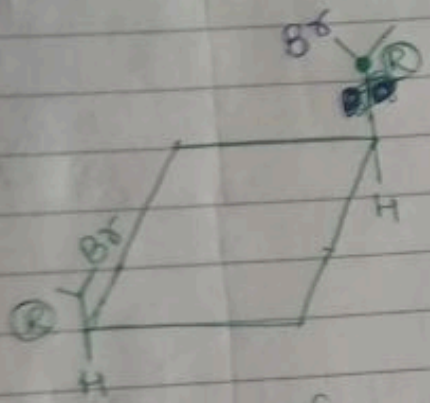
(Optical active)

9)



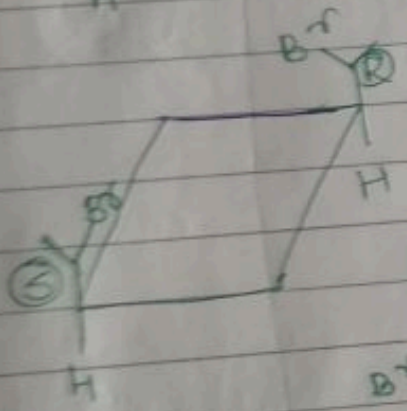
optical inactive

10)



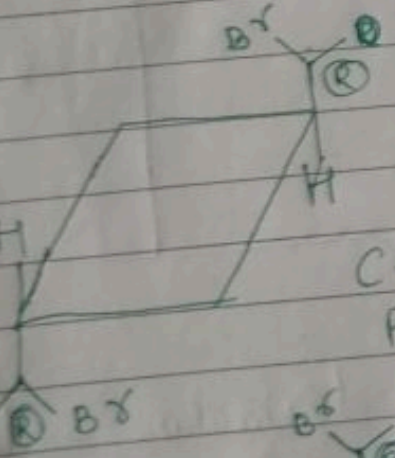
(optical active)

11)



(optical inactive)
(Meso compound)
(2 chiral carbon)

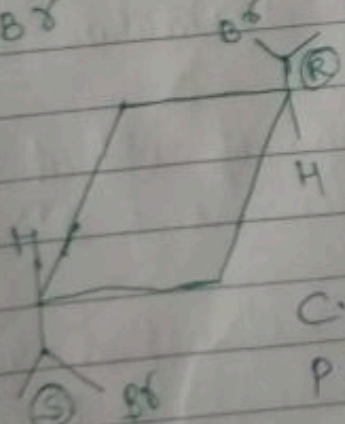
eg 12)



(optical active)

C·O·S = X
P·O·S = X

eg 13)



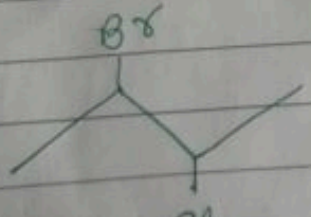
(optical inactive)
(Meso)

C·O·S = ✗ ✓
P·O·S = X

Calculation of no. of Stereoisomerism

n = no. of stereocentres	no. of optical active isomers	no. of optical inactive isomers
1) For dissimilar ends	2^n	zero
2) For similar ends n = even no. →	2^{n-1}	$2^{\frac{n}{2}} - 1$
3) For similar ends n = odd no. with Pseudo stereo centres	$2^{n-1} - 2^{\frac{n-1}{2}}$	$2^{\frac{n-1}{2}}$
4) For similar ends n = odd no. without Pseudo stereo centres	2^{n-1}	$2^{\frac{n-1}{2}}$

Eg 1)

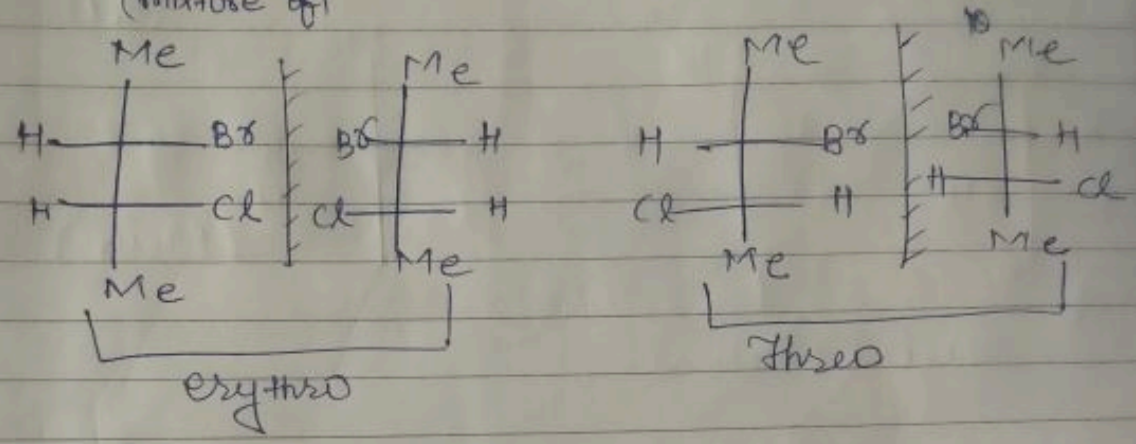


a) = dissimilar ends
b) n = 2
c) no. of optical active isomers = $2^n = 2^2 = 4$ (d,l ; d,l)
d) no. of optical inactive isomers = 0

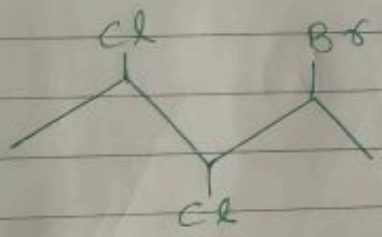
No. of pair of enantiomers = 2

5) No. of fractions on fractional distillation of all stereoisomers = 2

(mixture of)



eg 2)



⇒ dissimilar ends

$n = 3$

No. of optical active isomers

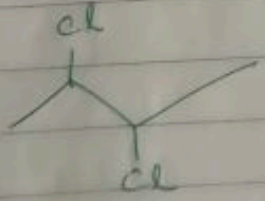
$2^n = 2^3 = 8$

⇒ No. of optical inactive isomers = 0

⇒ pair of enantiomers = 4

⇒ No. of fractions on fractional distillation = 4

eg 3)



⇒ similar ends

⇒ $n = 2$

⇒ no. of optical active isomers

$= 2^{n-1} = 2$ (d, l)

⇒ No. of optical inactive isomers

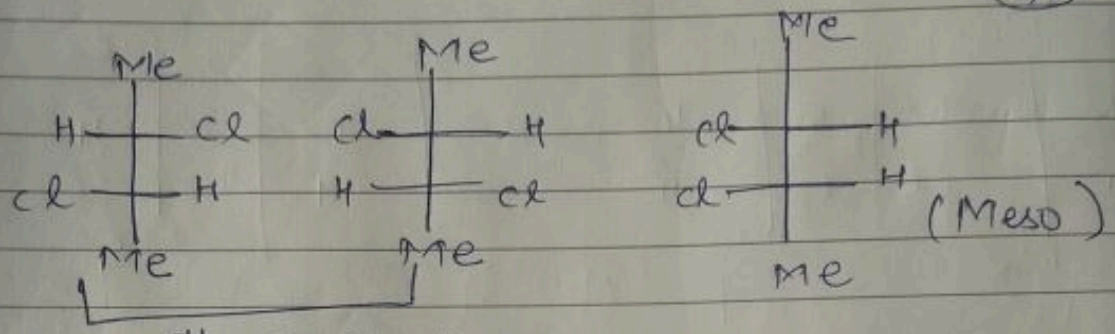
$= 2^{n/2 - 1} = 1$ (meso)

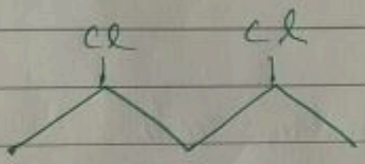
⇒ Total stereoisomers = 3

No. of pairs of enantiomers = 1

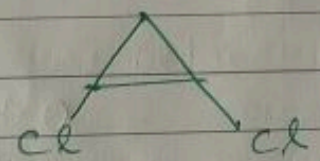
No. of fractions on fractional distillation = 2

(d, l) (Meso)

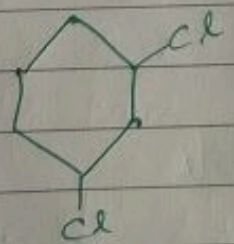


eg 4) 

n = 2
(Similar ends)

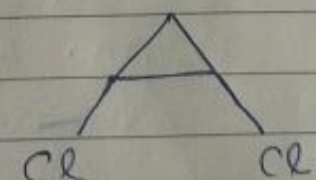
eg 5) 

n = 2
(Similar ends)

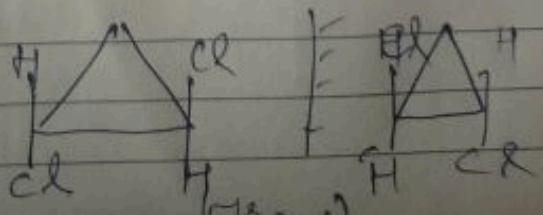
eg 6) 

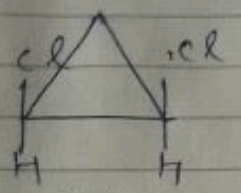
n = 2
(Similar ends)

eg. 3 (same)

Special for eg 5) 

Optical active = 2
Optical inactive = 1
(Meso)

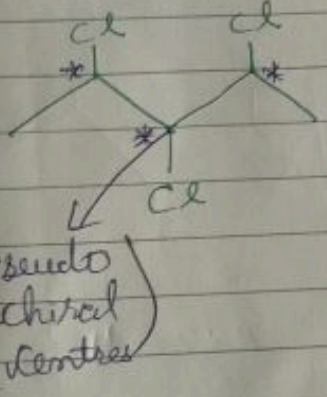




Cis
(Meso)

⇒ जब geometrical centres और chiral centres same आए तो ~~दोनों को अलग~~ सिर्फ chiral centres count करने से काम चल जाएगा

eg 7)



⇒ Similar ends
⇒ n = 3
⇒ No. of optical active ~~iso~~ iso

$= 2^{n-1} - 2^{\frac{n-1}{2}}$

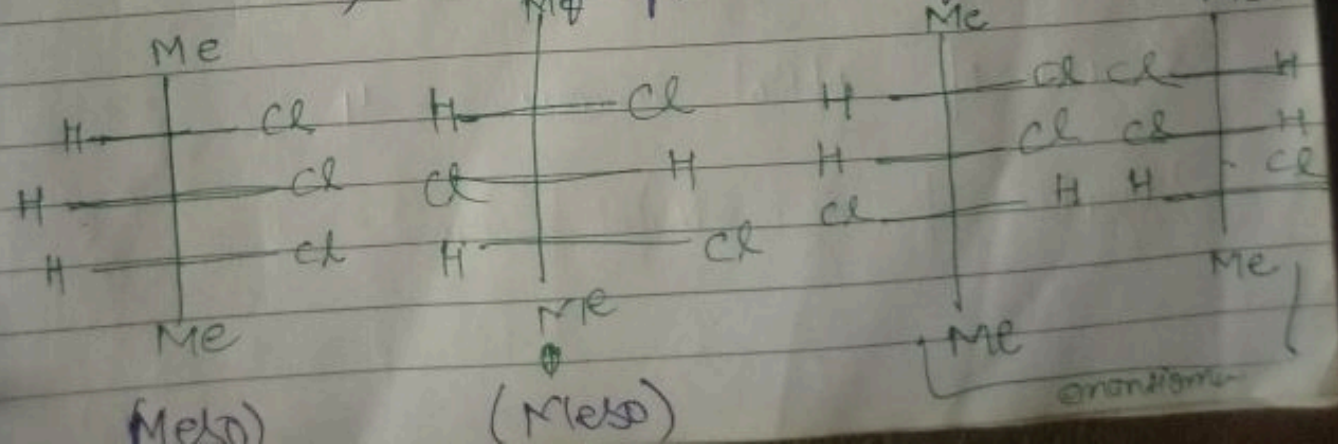
$= 2^2 - 2^1 = 2$ (d, l)

No. of optical inactive $= 2^{\frac{n-1}{2}}$

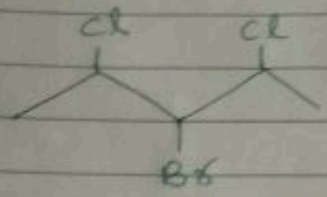
$= 2$

(Meso), (Meso)

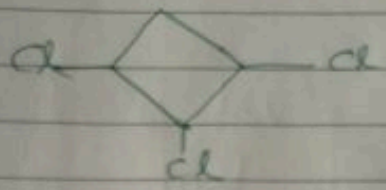
⇒ Total No. Stereoisomers = 4
⇒ pairs of enantiomers = 1
⇒ No. of fractions on frac. dist = 3



eg 8)

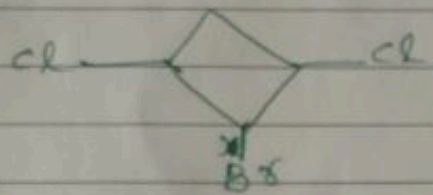


eg 9)

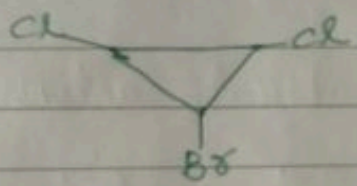


same as eg 7

eg 10)



eg 11)



eg 12) $CH_3-CH=CH-\underset{\underset{Br}{|}}{CH}-CH_3$ $\quad 2\times2=4$

1) ~~Cis -R~~ } enantiomer
2) Cis -S }
3) ~~Trans -R~~ } enantiomer
4) Trans - S }

$(n=2)$

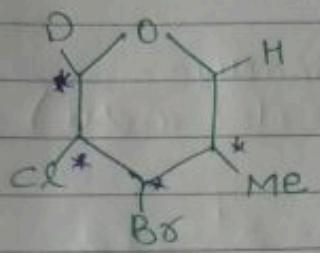
⇒ No. of stereoisomers = $2^2 = 4$

⇒ No. of optical isomers = 4

⇒ No. of geometrical isomer = 2

Note: जब No. of optical isomer पूछे तब n में geometrical centre भी count होंगे पर जब No. of geo isomer पूछे तब n में optical centre count नहीं होंगे।

eg 13)



$n = 4$

$2^4 = 16$ (Stereoisomers)

$2^4 = 16$ (optical ~~isomer~~active isomer)

(geometrical isomer)

eg 14) $CH_3-\underset{G.I}{CH=CH}-\underset{O.I}{CH=C=CH}-\underset{G.I}{CH=C=C=CH}-\overset{*}{\underset{Cl}{CH}}-CH_3$

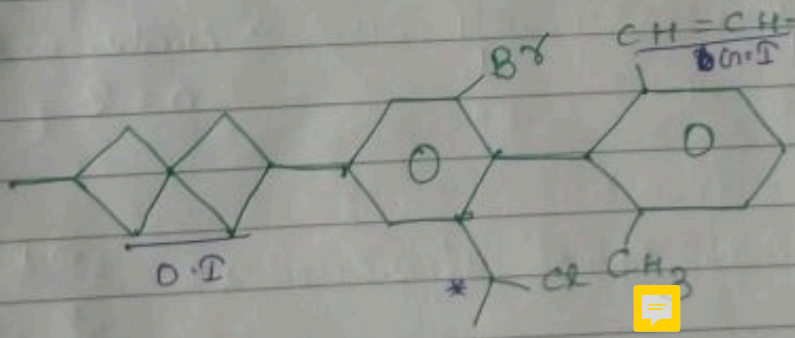
$n = 4$

optical active isomer $= 2^4 = 16$

Total Stereoisomer =

~~(optical active) =~~

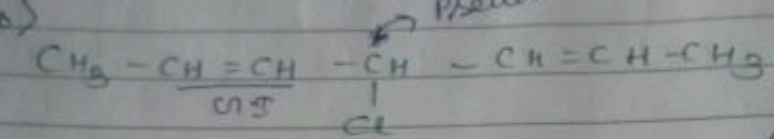
eg 15)



$n = 4$

Total Stereoisomers $= 2^4 = 16$

eg 16)



$n = 3$ (similar ends)

Cis - R - Trans } optical active
Cis - S - Trans
Cis - - - cis } optical inactive (Not meso)
Trans - - - Trans

Optical active $= 2^{n-1} - 2^{\frac{n-1}{2}} = 2^2 - 2^1 = 2$

Optical inactive $= 2^{\frac{n-1}{2}} = 2$

Total stereoisomers = 4

eg 17) $CH_3 - \underset{\underset{Br}{|}}{CH} - CH = CH - \underset{\underset{Br}{|}}{CH} - CH_3$ → similar ends $n = 3$

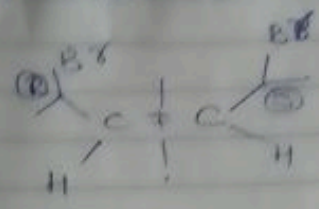
Total no. of optical active isomers $= 2^{n-1} = 2^2 = 4$

Total no. of optical inactive isomers $= 2^{\frac{n-1}{2}} = 2$

Total Isomers = 6

1) R - Cis - R
2) S - Cis - S
3) R - Cis - S (Meso)
4) R - Trans - R
5) S - Trans - S
6) R - Trans - S (Meso)

3) R - ch - S



COS ✓

POS : ✓

Q9 18) $CH_3-CH=C=CH-CH_3$ (Br, Br, Br, Br) → Pseudo geometrical centres

CH_3

→ n = 3

→ Similar ends

→ optical active isomers = $2^{n-1} - 2^{\frac{n-1}{2}} = 4 - 2 = 2$

→ optical inactive isomers = $2^{\frac{n-1}{2}} = 2$

→ total isomers = 4

1) R —— R] optical active
2) S —— S]
3) R — E — S] optical inactive
4) R — Z — S]

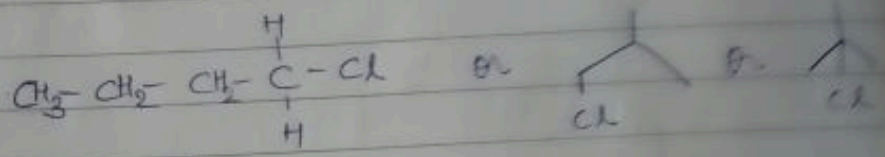
Q. 9 done

C · H

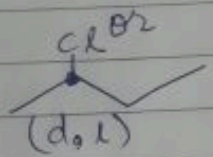
Q. Identify all possible compound for given molecular formula.

1) C_4H_9Cl
2) $C_5H_{11}Cl$
3) dichlorobutane
4) Trichlorocyclobutane
5) Dibromochlorocyclobutane

1)

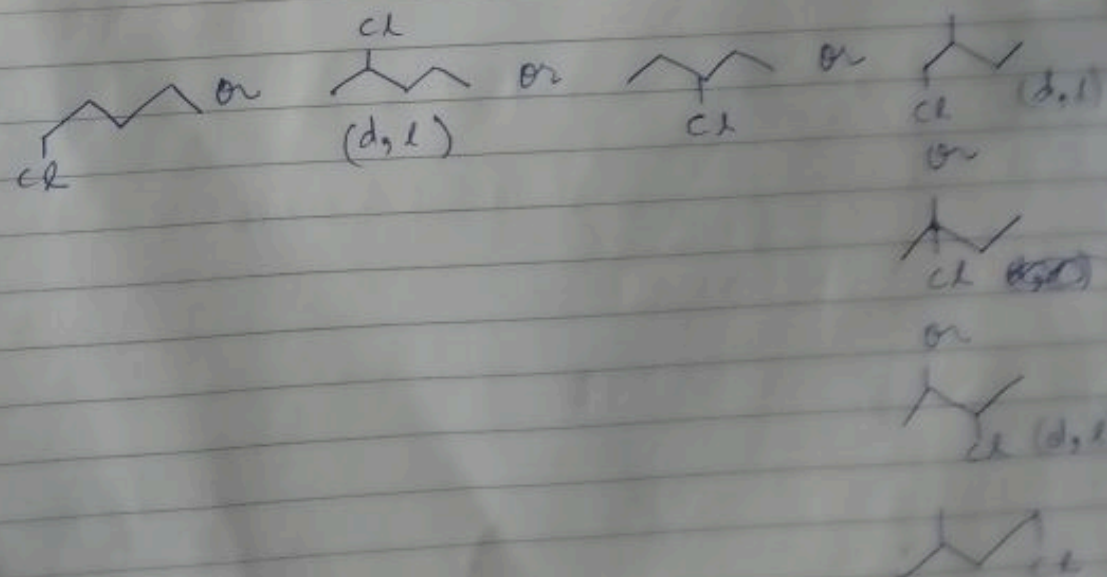


D.U = 0



(d, l) Total Stereoisomers = 5

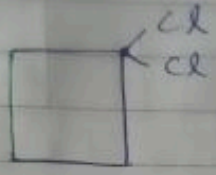
2) $C_5H_{11}Cl$ DU = 0

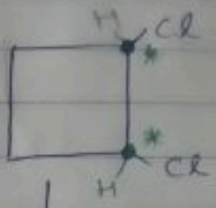




Total = 11 Ans

3)







(Cis + Trans)

$n = 2$

similar end

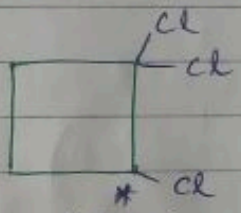
optical active $= 2^{n-1} = 2^1 = 2$

Optical inactive $= \frac{2}{2}^{\frac{n}{2}-1} = 1$

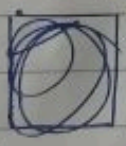
Total = 3 (d, l, meso)

= 6

4)

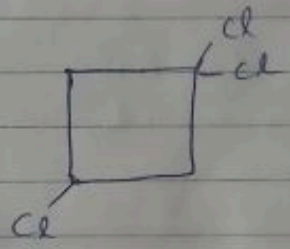


(d, l)

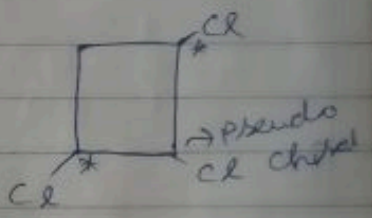


(2)

8



(1)



→ pseudo chiral

$n = 3$

(similar ends)

optical active $= 2^{n-1} - 2^{\frac{n-1}{2}}$

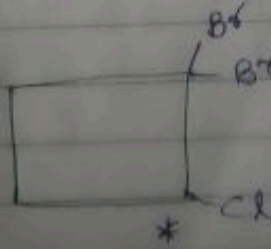
$= 4 - 2 = 2$

optical inactive = 2

Total = 4

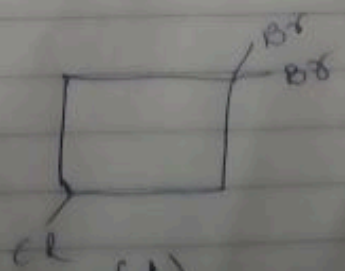
Total = 7

5)

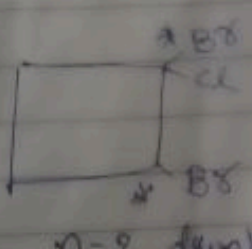


(d, l)

(2)

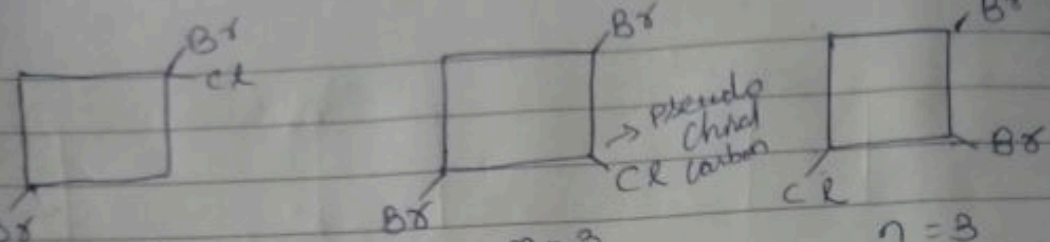


(1)



$n = 2$ dissimilar ends

$= 2^2 = 4$



O.A $2^{n-1} - 2^{\frac{n-1}{2}}$

$= 2$

O.I $= 2$

$= (4)$

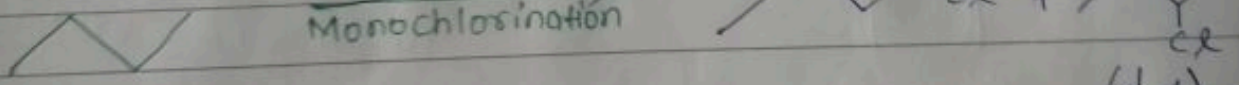
Total = 21

Q. Identify possible monochloro products

Hint

$$H-\overset{|}{\underset{|}{C}}-H \xrightarrow[-HCl]{Cl_2 + h\nu} -\overset{|}{\underset{|}{C}}-Cl$$

(Monochloro product)

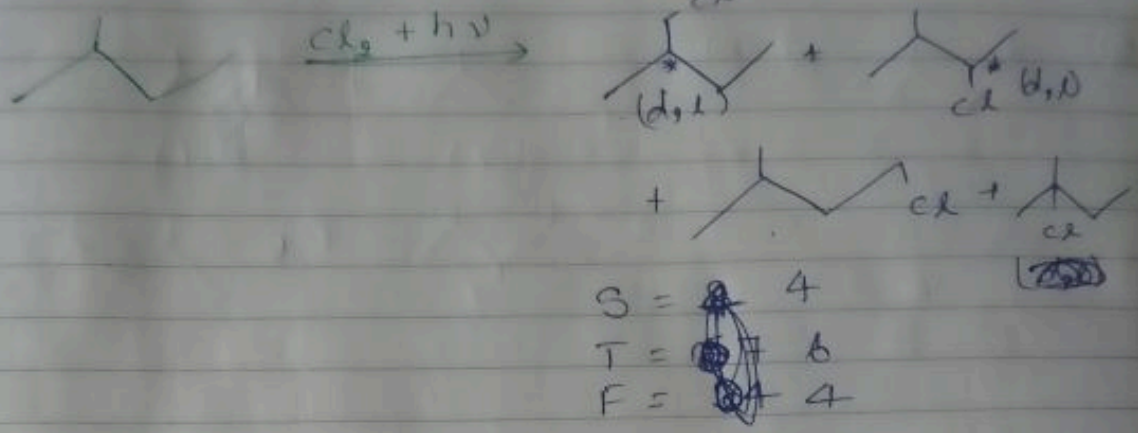


Structural products = S = 2

T = Total products = 3

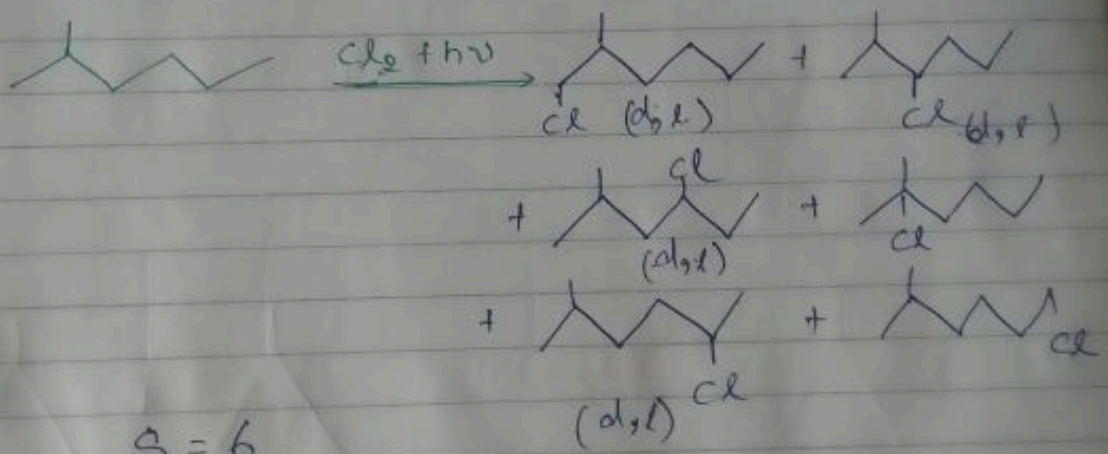
F = Total no. of fractions on fractional distillation of mix. of product = 2

2)



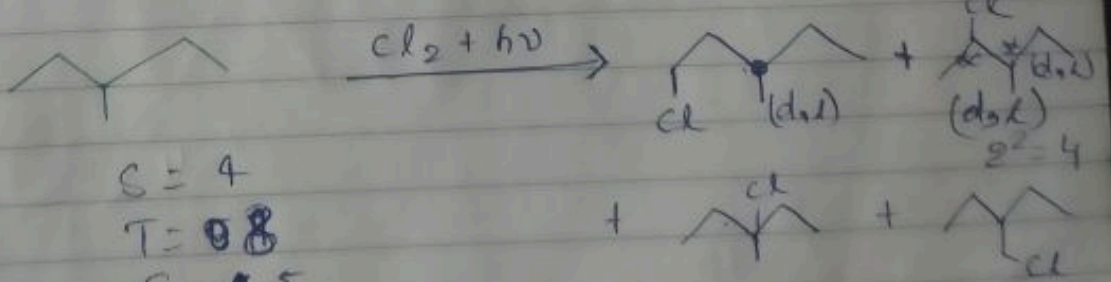
S = 4
T = 6
F = 4

3)



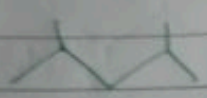
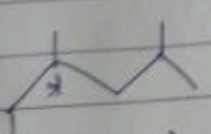
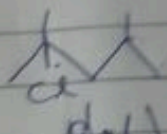
S = 6
T = 10
F = 6

4)

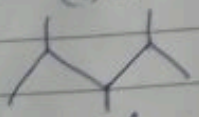


S = 4
T = 08
F = 05

b)

5)  $Cl_2 + h\nu$ →  + 

Cl (d, l) (d, l)

+  Cl D

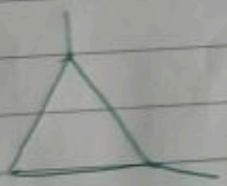
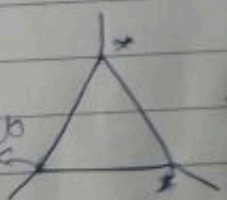
S = 3
T = 5
F = 3

6) $Cl_2 + h\nu$ →  +  + 

Cl

$2^2 = 4$

S = 3
T = 6
F = 4

7)  $Cl_2 + h\nu$ → Pseudo Optical centre Cl  + 

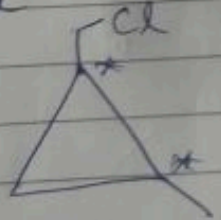
(3, 8) (4)

2^2

n = 3

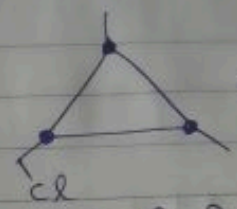
O.A = $2^{n-1} - 2\frac{n-1}{2} = 2$

O.I = $2^{\frac{n-1}{2}} = 2$



$2^2 = 4$

S = 3
T = 12
F = 7

8)  $Cl_2 + h\nu$ 



$n = 3$

$n = 3$

O.A = 2

O.I = 2

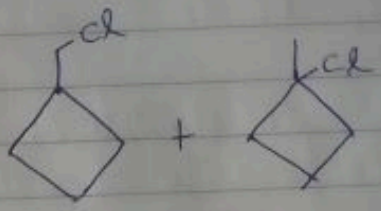
$O.A = 2^{n-1} - 2^{\frac{n-1}{2}} = 2$

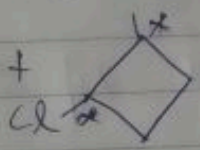
$O.I = 2^{\frac{n-1}{2}} = 2$

S = 2

T = 8

F = 6

9)  $Cl_2 + h\nu$ 



$2^2 = 4$

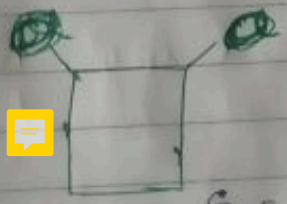


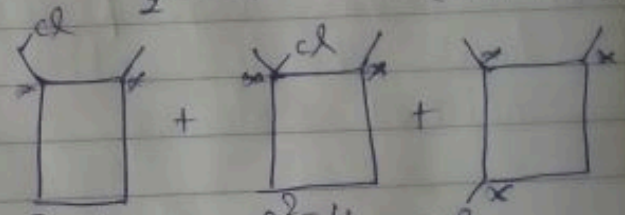
(Cis + trans)

S = 4

T = 8

F = 6

10) 

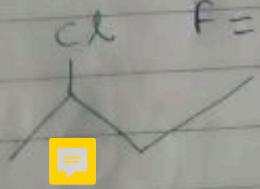


$2^2 = 4$ $2^2 = 4$

S = 3

T = 6

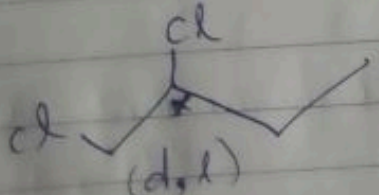
F = 8

11) 

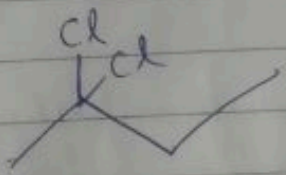


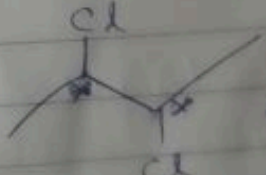
n = 3

$2^3 = 8$



(d,l)





n = 2

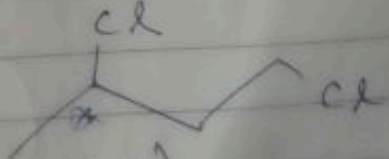
$OA = 2^{n-1} = 2$

$OI = 2^0 = 1$

S = 4

T = 8

F = 5



(l,l)